

Discrete Initial Groups and Omnific Normalization

Sign-tree surgery, a sharp image classification,
an Ehrlich–Kaplan question, and convex factors

AI-assisted research draft prepared for the Surreal project

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Abstract

We give an explicit construction proposed as an affirmative solution of Ehrlich and Kaplan’s question whether every discretely ordered initial subgroup of the surreal numbers is order-isomorphic to an initial subgroup of the omnific integers (Question 2 of the arXiv preprint, Question 9.1 of the journal version). The construction is not ordinary multiplication by the reciprocal of the least positive element. Instead, a sign-sequence operation moves the minimum exponent to zero, while preserving all right-ancestor relations away from that minimum; a separate rescaling changes only the bottom coefficient group to the ordinary integers. The Ehrlich–Kaplan characterization of initial Hahn subgroups then establishes initiality of the image.

The argument yields more than existence. For a prescribed least positive element $2^{-n}\omega^{-\alpha}$, we characterize the images exactly by containment of an explicit dyadic-spine subgroup $K_\alpha \subseteq \mathbf{Oz}$. We identify the smallest and largest initial groups with that least positive element, give sharp cardinality consequences, construct an initial realization of the quotient by the least positive cyclic subgroup, and extend the additive normalization to rectangular surcomplex modules. All normal-form transports preserve leading truncations and set-indexed strong summability. Counterexamples explain why scalar normalization, exponent translation, and unrestricted inversion do not suffice.

A second manuscript realizes every convex subquotient of an initially realizable group by sign-tree compression, and shows that a Presburger group is initially realizable exactly when its residues in $\widehat{\mathbb{Z}}$ are ordinary integers.

Status and scope. This is an unrefereed research manuscript with full proposed proofs, not an independently certified resolution or a claim of established priority. The principal external mathematical input is explicitly identified. Finite regression checks accompany the text; they do not certify the transfinite argument. No new Lean formalization is claimed. A targeted literature and repository review found the cited question and its motivating example, but did not establish exhaustive novelty. At placement in the collection every step was checked by hand and no gap was found (section 19.5); that check is not a refereeing, and the literature search was targeted, not exhaustive. The second manuscript (section 1.5) has the same status.

Keywords: surreal numbers; omnific integers; initial subgroups; sign sequences; Hahn series; ordered abelian groups.

Mathematical focus: additive and simplicity-theoretic structure, not multiplicative or exponential equivalence.

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1 The problem and the main results

1.1 A specific published question

An additive subgroup of **No** is *initial* when it contains every proper sign-sequence prefix of each of its members. This is a much stronger requirement than being an ordered subgroup. An initial subgroup is *discrete* when it is nonzero and has a least positive element. Throughout, “isomorphism” of ordered groups preserves addition and order; it need not preserve the ambient simplicity relation.

Ehrlich and Kaplan ask:

“Is every discrete initial subgroup of **No** isomorphic to an initial subgroup of **Oz**?”

This is Question 2 in Section 9 of the explicitly versioned preprint [1], printed page 18. The journal article appeared in 2018 [2]. We use the preprint’s numbering throughout to avoid mixing versions. The question also appears in Kaplan’s thesis [3].

In the journal version the same question is Question 9.1, with the same wording and the same motivating example. The results of [1] used below are numbered differently there; the correspondence, checked when this report was placed in the collection against the author-hosted text of the journal version [2], is:

arXiv v1 [1]	journal version [2]	use here
Lemma 3	Lemma 5.1	initial subgroups of $\mathbb{R}$, imported fact 2.3
Theorem 1	Theorem 5.1	the initiality criterion, imported fact 2.2
Proposition 9	Proposition 5.2	shape of the least positive element
Question 2	Question 9.1	the question addressed here
Question 3	Question 9.2	the initial-monoid problem, not addressed
Theorem 6	Theorem 8.1	optimality with class models (source 02)
Question 1	Question 8.1	set-model optimality, not answered

Below, the arXiv numbers are used, except where the wording of the journal version matters (after imported fact 2.2).

The distinction between a group and a ring matters. A discrete initial subgroup can contain $1/2$ or a nonzero infinitesimal, whereas an omnific integer has neither a nonintegral constant coefficient nor a negative normal-form exponent. The task is therefore to change the initial *realization* of the ordered group, not to prove that its original realization already lies in **Oz**.

The paper addresses precisely this existence question and several structural refinements. It does not claim a classification of all ordered abelian groups admitting initial embeddings, nor a solution of the separate initial-monoid problem (Question 3 of [1], Question 9.2 of the journal version).

1.2 Normalization and its exact range

Write $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{D} = \mathbb{Z}[1/2]$. If α is an ordinal, let

$$p_\alpha = -\alpha = \langle - \rangle^\alpha, \quad q_\beta = \langle + \rangle \frown \langle - \rangle^\beta \quad (\beta \in \mathbf{On}).$$

Here $\frown$ means concatenation of sign sequences, not surreal addition. For example, $q_0 = 1$, $q_1 = 1/2$, and $q_m = 2^{-m}$ for finite m . Define the finite-support additive subgroup

$$K_\alpha = \mathbb{Z} \oplus \bigoplus_{\beta < \alpha} \mathbb{D} \omega^{q_\beta} \subseteq \mathbf{Oz}. \quad (1)$$

The direct sum means that each element has only finitely many nonzero summands. In particular, $K_0 = \mathbb{Z}$.

Theorem A (Omnific normalization). Let $A \subseteq \mathbf{No}$ be a nonzero discrete initial additive subgroup. There are unique $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$ such that its least positive element is

$$\varepsilon = 2^{-n}\omega^{-\alpha}.$$

An explicitly defined map $N_{\alpha,n}$ restricts to an order-preserving group isomorphism

$$N_{\alpha,n} : A \longrightarrow H, \quad H \subseteq \mathbf{Oz} \text{ initial}, \quad N_{\alpha,n}(\varepsilon) = 1.$$

The map extends to a real-linear order-isomorphism between the full normal-form spaces supported in $[-\alpha, \infty)$ and $[0, \infty)$. This extension preserves support order types, all leading truncations, and set-indexed strong summability in both directions.

Theorem B (Sharp image classification). Fix $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$. Under the same explicit map $N_{\alpha,n}$, the initial groups with least positive element $2^{-n}\omega^{-\alpha}$ correspond bijectively, with inclusions preserved and reflected, to the initial groups $H \subseteq \mathbf{Oz}$ satisfying

$$K_\alpha \subseteq H.$$

The inverse is explicit. Thus the dyadic-spine condition is both necessary and sufficient, not merely an artifact of a sufficient construction.

Theorem C (Removing the bottom cyclic layer). For every group A in [Theorem A](#), the subgroup $\mathbb{Z}\varepsilon$ is convex, and the ordered quotient $A/\mathbb{Z}\varepsilon$ has an explicit initial realization in $\mathbf{No}$. The corresponding short exact sequence splits as an additive group, with a canonical splitting relative to the given Conway normal forms. Equivalently,

$$A \cong B \overrightarrow{\times} \mathbb{Z}$$

for an explicitly constructed initial subgroup $B \subseteq \mathbf{No}$, where the first coordinate is dominant in the lexicographic order.

The main construction is in [sections 4 and 5](#); the proofs of the three theorems are completed in [sections 6 to 8](#), respectively. [Theorem B](#) contains more information than [Theorem A](#): it identifies exactly which normalized realizations can be moved to any prescribed least positive scale by this construction.

1.3 Why a change of exponents is necessary

Multiplication by ε^{-1} certainly gives an ordered-group isomorphism. Initiality, however, refers to sign prefixes rather than numerical intervals, and multiplication does not generally preserve it. Even $A = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\omega$ is an initial group for which $2A = \mathbb{Z} + 2\mathbb{Z}\omega$ is not initial: 2ω belongs to the image but its simpler predecessor ω does not.

Our operation treats two different issues separately. First, it relocates the minimum of the *exponent tree* to zero. Second, it rescales only the real coefficient at that minimum. No claim of multiplicativity is made, and none is needed for the published ordered-group question.

1.4 Conventions shared with the collection

This report is built from two manuscripts: source 01, the normalization manuscript ([sections 1 to 10](#) and [Appendix A](#)), and source 02, the convex-factor manuscript ([section 1.5](#), [sections 11 to 18](#) and [Appendix C](#)). Their provenance, the placement reviews and the relation to other reports are in [section 19.5](#). Statements added in the merge of the two sources, and proved here, are marked [merge]. The following conventions are fixed here once; the notation of source 02 is fixed in [section 11.1](#).

1. *Initial* always means closed under proper sign-sequence prefixes, as in the definition of the definable-surreals report (dsn:sec:foundations). It never means an initial segment of the numerical order, and it has nothing to do with the *initial forms* (leading forms of polynomial solutions) of the Diophantine report (odg:prop:initial).
2. $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, with Π the class of purely infinite normal forms, is the omnific ring of the collection (odg:def:rings, osq:eq:Ozdef, found:eq:omnific); (4) below is the same description. For $x \in \mathbf{Oz}$, $\text{ct}(x)$ is its coefficient at exponent 0, as there. Only the additive and order structure of $\mathbf{Oz}$ is used; its ring structure plays no role.
3. The symbols Θ_α , Ψ_α , q_β and $N_{\alpha,n}$ are local to this report. They are unrelated to the theta series and to the nome or dilation parameter q of other reports.

1.5 Convex factors and profinite obstructions

Theorem C removes the least positive cyclic subgroup. The second source of this report, *Convex Factors and Profinite Obstructions for Initial Surreal Groups*, removes an arbitrary convex subgroup. Call an ordered abelian group *initially realizable* if it is order-isomorphic to an additive subgroup of $\mathbf{No}$ closed under all proper sign-sequence prefixes, and write $G \in \text{Init}$ for this property; the zero group is included. A *convex factor* of G is C'/C , where $C \subseteq C'$ are convex subgroups of G and the quotient has its usual ordered group structure.

Theorem D (Convex-factor closure). If $G \in \text{Init}$ and $C \subseteq C'$ are convex subgroups of G , then $C'/C \in \text{Init}$. For an actual initial realization $G \subseteq \mathbf{No}$, the construction is explicit: restrict normal forms to the corresponding convex exponent interval and compress its sign tree.

Moreover, every convex subgroup sequence

$$0 \longrightarrow C \longrightarrow G \longrightarrow G/C \longrightarrow 0$$

splits as a sequence of abelian groups. In the chosen normal-form realization the section is order-preserving and the decomposition is lexicographic. The coordinate projections and compressed factor realizations satisfy precise composition laws under nested restrictions.

The theorem concerns *initial realizability*, not literal initiality of the original subgroup C or of a selected complement. It also does not assert a splitting invariant under every abstract automorphism. The explicit splitting is attached to a chosen initial realization. With $C' = G$ it answers question 20.1: every convex subgroup of an initial group has an explicit initial realization of the quotient by a normal-form transport, and for $C = \mathbb{Z}e$ that transport is the construction of section 8 (proposition 13.5). It also answers question 20.2: no extra hypothesis is needed at a limit stage (corollary 13.6).

A $\mathbb{Z}$ -group is a discretely ordered abelian group G with least positive element e such that $G/\mathbb{Z}e$ is divisible. These are exactly the ordered-group models of additive Presburger arithmetic [10]. Their coherent standard residues define

$$\text{res}_G : G \longrightarrow \widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z}, \quad \text{res}_G(e) = 1.$$

Here $\mathbb{Z}$ on the right is the ordinary, diagonally embedded copy of the integers, not a nonstandard integer substructure.

Theorem E (Presburger initiality criterion). For a set-sized $\mathbb{Z}$ -group G , the following are equivalent:

$$\begin{aligned} G \in \text{Init}; \quad & G \text{ has an initial realization in } \mathbf{Oz}; \quad & \mathbb{Z}e \text{ is a direct summand of } G; \\ \text{res}_G(G) = \mathbb{Z}; \quad & G \cong G^{\text{div}} \overrightarrow{\times} \mathbb{Z}, \quad & G^{\text{div}} = \bigcap_{n \geq 1} nG. \end{aligned}$$

The group G^{div} is divisible. In the split case its complement to $\mathbb{Z}e$ is unique.

This is a global compatibility condition, not a condition on any one finite modulus. Every $\mathbb{Z}$ -group has the same individual quotients $G/nG \cong \mathbb{Z}/n\mathbb{Z}$. Initiality distinguishes whether the residues of a *single element*, across all standard moduli, come from one ordinary integer.

Theorem F (Explicit countermodels). There are $2^{\aleph_0}$ pairwise nonisomorphic countable rigid $\mathbb{Z}$ -groups which are not initially realizable, although they are all elementarily equivalent to $\mathbb{Z}$. There are also countable densely ordered groups, elementarily equivalent to an explicit initial subgroup of **No**, which are not initially realizable.

No recursively saturated $\mathbb{Z}$ -group, and no nonprincipal ultrapower over $\mathbb{N}$ of a $\mathbb{Z}$ -group, is initially realizable. The ordered additive group completion of any nonstandard model of Peano arithmetic is likewise not initially realizable.

The proofs are in section 13 (Theorem D), section 14 (Theorem E) and section 16 (Theorem F). In source 02 these three theorems are Theorems A, B and C; they are lettered D, E and F here because source 01's Theorems A–C keep their letters.

What is new in source 02, and what is not. The primary external structural input is again the Ehrlich–Kaplan characterization of initial Hahn subgroups ([imported fact 2.2](#)). Their treatment of convex subgroups asks which subgroups in the *given* realization are themselves initial. The construction of source 02 changes the realization of a factor and preserves its ordered additive structure. Source 01 already gives the normalization of [Theorem A](#), the integer-bottom suspension and the relative splitting criterion for discrete groups ([theorem 8.5](#) and [corollary 8.6](#)); its [section 8](#) explicitly leaves arbitrary convex quotients outside its conclusions. Source 02 does not claim those results again as new. It replaces the special positive-root deletion by a compression operation valid for arbitrary convex exponent intervals; in particular it handles cuts with no endpoint and convex factors not accessible by a finite iteration of least-positive quotients.

The profinite residue map and its divisible kernel belong to the classical structure theory of $\mathbb{Z}$ -groups; see especially [10, Sections 2 and 4]. Likewise, the extension parameter $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ is classical [11, Example 8.27]. Source 02's arithmetic contribution is their explicit connection with initial surreal realizability, together with the displayed constructions and obstructions. The existence of rigid Presburger models is not claimed as a new phenomenon. The results are not announced as a resolution of the set-model version of Ehrlich–Kaplan's optimality question (Question 1 of [1], Question 8.1 of the journal version); a precise comparison is in [section 18.2](#). A targeted review found no matching formulation of the main convex-factor theorem, but this is not an exhaustive priority search.

2 Foundations and the imported initiality criterion

2.1 Sets, classes, signs, and two orders

We work in NBG with global choice, in the conventional class-sized formulation of surreal numbers. Every individual surreal is a set-length sign sequence. Every individual normal form has set support. All strongly summable families considered below are indexed by sets. Statements about families of class-sized groups are understood as correspondence statements about class parameters, not as assertions that a class of all classes exists.

For sign sequences u, v , write $u \prec v$ for a proper prefix and $u \preceq v$ for a prefix allowing equality. Their ordinary numerical order is determined at the first disagreement, with

$$- < \text{undefined} < +.$$

The empty sequence is the number 0. Put

$$\text{Pred}(x) = \{y : y \prec x\}, \quad \text{R}_s(x) = \{y \prec x : x < y\}, \quad \text{L}_s(x) = \{y \prec x : y < x\}.$$

In sign language, a prefix obtained by stopping just before a minus sign is a right ancestor; stopping just before a plus sign gives a left ancestor. Each of these ancestor collections is a set.

A subclass $E \subseteq \mathbf{No}$ is initial precisely when $\text{Pred}(x) \subseteq E$ for every $x \in E$. This does not mean that E is downward closed in the numerical order. For instance, $\mathbf{No}_{\geq 0}$ is initial as a sign tree but is a numerical final segment.

If u and v have lengths λ and μ , then $u \frown v$ has length $\lambda +_{\text{ord}} \mu$. All additions of *lengths* in this paper are ordinal additions. Concatenation is associative and preserves comparison between tails sharing the same prefix. At a limit ordinal there is no last sign to remove: “remove the first sign” or “remove a specified concatenated block” always means restricting to the appropriate interval and taking its order type. These conventions agree with the sign-sequence framework of [5].

2.2 Hahn spaces without an exponent group

For an ordered subclass $E \subseteq \mathbf{No}$, denote by $\mathcal{H}_{\mathbb{R}}(E)$ the additive space of formal expressions

$$x = \sum_{\gamma \in S} r_{\gamma} \omega^{\gamma}, \quad r_{\gamma} \in \mathbb{R} \setminus \{0\}, \quad (2)$$

where $S \subseteq E$ is a set reverse-well-ordered by $<$. Conway normal forms identify these expressions with the corresponding surreal numbers. The ordering is the sign of the coefficient at the greatest exponent of a nonzero difference. For $x \neq 0$, write

$$\deg_H(x) = \max \text{supp}(x), \quad \text{lc}(x) = r_{\deg_H(x)}.$$

We do not assert that E is an additive group. Accordingly, $\mathcal{H}_{\mathbb{R}}(E)$ is used here as an additive, real-linear Hahn space, not necessarily as a ring. This distinction permits nonadditive exponent reindexings.

Suppose $S = \{\gamma_{\xi} : \xi < \lambda\}$ is enumerated in strictly decreasing order. A leading truncation of x is

$$x_{<\eta} = \sum_{\xi < \eta} r_{\gamma_{\xi}} \omega^{\gamma_{\xi}} \quad (\eta \leq \lambda).$$

“Truncation closed” will always refer to these leading truncations, including those at limit indices, not to closure under arbitrary subseries. The normal-form and monomial facts used here are part of the standard theory summarized in [1]; we never choose a noncanonical Hahn embedding of an arbitrary ordered group.

Definition 2.1 (Strong summability). A set-indexed family $(x_i)_{i \in I}$ in a Hahn space is strongly summable if $\bigcup_i \text{supp}(x_i)$ is reverse-well-ordered and each exponent occurs in only finitely many supports. Its sum is the normal form obtained by the finite coefficient sum at each exponent, with zero coefficients discarded.

For a subgroup $G \subseteq \mathcal{H}_{\mathbb{R}}(E)$, set

$$G_{\gamma} = \{r \in \mathbb{R} : r\omega^{\gamma} \in G\}.$$

It is *cross-sectional on E* when $\omega^{\gamma} \in G$ for every $\gamma \in E$.

Imported fact 2.2 (Ehrlich–Kaplan, concrete form). *An initial subgroup of $\mathbf{No}$ is a truncation-closed, cross-sectional Hahn subgroup on its initial exponent class $\Gamma(G) = \{\gamma : \omega^{\gamma} \in G\}$. Its nonzero coefficient groups are initial subgroups of $\mathbb{R}$, and*

$$\gamma, \eta \in \Gamma(G), \quad \eta \in R_s(\gamma) \implies \mathbb{D} \subseteq G_{\eta}. \quad (3)$$

Conversely, these conditions imply that the canonical normal-form image is initial in $\mathbf{No}$.

This is the form of [1, Theorem 1 and its proof] needed here. The last sentence is important: the sufficiency proof establishes initiality of the *specified canonical image*, not just existence of some other initial embedding.

The journal version states this concrete form outright. Its Theorem 5.1, the counterpart of [1, Theorem 1], says that a subgroup of **No** is initial if and only if it is isomorphic “via the canonical isomorphism” to a truncation-closed, cross-sectional Hahn subgroup whose exponent class is an initial ordered subclass of **No** and whose coefficient groups satisfy the two conditions above [2]. The canonical isomorphism is the normal-form identification of **No** with the Hahn group, so the journal statement is *imported fact 2.2* as used here. The criterion is imported as a published statement. Neither the source manuscript nor the placement review re-proves it (section 19.5).

Imported fact 2.3 (Real coefficient groups). *A nonzero initial additive subgroup of $\mathbb{R}$ either contains $\mathbb{D}$ or is $2^{-m}\mathbb{Z}$ for an integer $m \geq 0$ [1, Lemma 3] (Lemma 5.1 of the journal version). In particular, a discrete one has least positive element 2^{-m} .*

No proof in the present paper replaces *imported fact 2.2* by the weaker assertion that initiality of the exponent set alone is sufficient. The coefficient groups and the right-ancestor condition are essential.

2.3 Omnific integers and coefficient isolation

We use the normal-form description

$$\mathbf{Oz} = \left\{ \sum_{\gamma > 0} r_{\gamma} \omega^{\gamma} + m : m \in \mathbb{Z} \right\}, \quad (4)$$

with set reverse-well-ordered positive support and arbitrary real coefficients. This is Conway’s canonical integer part, as recalled in [3, Section 6.2], and the collection’s $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ (section 1.4). It is an initial additive subgroup: apply *imported fact 2.2* to $E = \mathbf{No}_{\geq 0}$, coefficient group $\mathbb{Z}$ at zero and $\mathbb{R}$ at every positive exponent. Zero is never a right ancestor of a nonnegative number, so the exceptional coefficient group causes no violation of (3). Every positive element of $\mathbf{Oz}$ is at least 1; this discreteness is also *odg:prop:units* of the Diophantine report.

Lemma 2.4 (Isolating a normal-form term). *Let G be an additive subgroup closed under leading truncations. If $x \in G$ has a nonzero coefficient r_{γ} at exponent γ , then $r_{\gamma} \omega^{\gamma} \in G$.*

Proof. If $\gamma = \gamma_{\eta}$ in the decreasing enumeration of the support, subtract the truncation at η from the truncation at $\eta + 1$. Both belong to G . This also works when η is a limit ordinal. $\square$

This lemma permits removing a specified individual term. It does not license an arbitrary infinite subseries of an element of G .

3 Locating the least positive scale

Lemma 3.1 (A minimum in an initial tree). *If a nonempty initial subclass $E \subseteq \mathbf{No}$ has a numerical minimum p , then $p = -\alpha$ for a unique ordinal α . Moreover, the cone $C_{\alpha} = \{x \in \mathbf{No} : x \geq -\alpha\}$ is initial.*

Proof. A plus sign in p would give a proper prefix smaller than p . Initiality would place that prefix in E , contradicting minimality. Thus every sign in p is minus.

For the second assertion, let $p = -\alpha$, $x \geq p$, and $v \prec x$. If $v < p$, then at the first disagreement with the all-minus sequence p one would have to find either a sign smaller than minus, which is impossible, or a minus sign after p has ended. The latter would force x itself to be below p . Hence $v \geq p$. $\square$

Proposition 3.2 (Bottom coefficient structure). *Let A be a nonzero discrete initial subgroup with least positive member ε . There are unique $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$ such that, writing $p = -\alpha$ and $c = 2^{-n}$,*

$$\varepsilon = c\omega^p, \quad \min \Gamma(A) = p, \quad A_p = c\mathbb{Z}. \quad (5)$$

Every support of every member of A lies in C_α .

Proof. By imported fact 2.2, A is truncation closed. Write the leading term of ε as $r\omega^p$ with $r > 0$. This term belongs to A . If the remaining tail t were nonzero, then $|t| \in A$ and $0 < |t| < \varepsilon$, since the leading exponent of t is strictly smaller than p . This contradicts the choice of ε . Consequently $\varepsilon = r\omega^p$.

The exponent p lies in $\Gamma(A)$ by the cross-sectional normal-form description. If $\gamma < p$ belonged to $\Gamma(A)$, then $0 < \omega^\gamma < r\omega^p = \varepsilon$, another contradiction. Thus p is the minimum of $\Gamma(A)$. By lemma 3.1, $p = -\alpha$ for a unique ordinal α .

The real coefficient group A_p has least positive element r . By imported fact 2.3, it is $2^{-n}\mathbb{Z}$ for a unique $n \in \mathbb{N}$, and $r = 2^{-n}$. All supports lie in $\Gamma(A)$, proving the last assertion. $\square$

The shape of the least positive element is consistent with [1, Proposition 9 and its proof] (Proposition 5.2 of the journal version); the argument above also records the support and coefficient data needed later.

Corollary 3.3 (Forced ancestors). *Under (5), for every $\beta < \alpha$ one has*

$$-\beta \in \Gamma(A), \quad \mathbb{D}\omega^{-\beta} \subseteq A.$$

Proof. The number $-\beta$ is a right ancestor of $-\alpha$. Initiality gives the exponent, and (3) gives all dyadic coefficients there. $\square$

The bottom coefficient group can be discrete, but all these ancestors must carry dyadically divisible coefficient groups. This is the information that survives normalization as the subgroup K_α .

4 Moving a minimum to zero in the sign tree

4.1 Definition and inverse

Fix $\alpha \in \mathbf{On}$, put $p = -\alpha$, and keep $C_\alpha = [p, \infty)$. Every $x \in C_\alpha$ belongs to exactly one of the following cases: $x = p$; $x = p \frown \langle + \rangle \frown u$ for a unique tail u ; or p is not a prefix of x . Indeed, an extension through $p \frown \langle - \rangle$ would lie below the cone.

Definition 4.1 (Root relocation). Define $\Theta_\alpha : C_\alpha \rightarrow \mathbf{No}_{\geq 0}$ by

$$\Theta_\alpha(x) = \begin{cases} 0, & x = p, \\ \langle + \rangle \frown p \frown u, & x = p \frown \langle + \rangle \frown u, \\ \langle + \rangle \frown x, & p \not\preceq x. \end{cases} \quad (6)$$

The inverse candidate $\Psi_\alpha : \mathbf{No}_{\geq 0} \rightarrow C_\alpha$ is

$$\Psi_\alpha(0) = p, \quad \Psi_\alpha(\langle + \rangle \frown v) = \begin{cases} p \frown \langle + \rangle \frown u, & v = p \frown u, \\ v, & p \not\preceq v. \end{cases} \quad (7)$$

The symbols in these formulas are sign blocks. For $\alpha = 0$, the block p is empty and both maps are the identity. For $\alpha > 0$, one may picture the operation as deleting the old minimum node, moving its right subtree into the vacated position, and adding a new common root. The formulas, not this picture, govern limit-length cases.

Lemma 4.2 (The inverse is well-defined). *The maps (6) and (7) are inverse bijections between C_α and $\mathbf{No}_{\geq 0}$.*

Proof. Every nonempty image under Θ_α begins with plus and is therefore positive. Conversely, let $y = \langle + \rangle \frown v > 0$. If $p \preccurlyeq v$, the specified preimage extends p through plus and is greater than p . If $p \not\preccurlyeq v$, then $v > p$: a sequence below an all-minus word must extend that entire word through minus, contrary to the assumption. Thus Ψ_α takes values in the cone.

If $x = p$, the composite $\Psi_\alpha \Theta_\alpha$ returns p . If $x = p \frown \langle + \rangle \frown u$, removing the first sign of $\langle + \rangle \frown p \frown u$ leaves $p \frown u$, and the first inverse case returns x . In the remaining case, removing the first sign leaves x , to which the second inverse case applies. The same two cases for v prove $\Theta_\alpha \Psi_\alpha(y) = y$ for positive y , and zero is immediate. Uniqueness of tails follows from the order types of the corresponding ordinal intervals, so none of these steps requires a successor length. $\square$

Lemma 4.3 (Order preservation). *The bijection Θ_α is strictly increasing.*

Proof. The minimum p maps to the minimum 0. On the block

$$U = \{p \frown \langle + \rangle \frown u : u \in \mathbf{No}\},$$

comparison is comparison of the tails u , before and after the map. On the remaining class $V = C_\alpha \setminus (U \cup \{p\})$, adding one common initial plus sign also preserves comparison.

It remains to compare the two blocks. Every member of U is smaller than every member v of V : before the end of p , such a v either stops or first has plus where p has minus. In both cases v is greater than every extension of p . After applying Θ_α , cancel the common initial plus sign. One then compares an extension $p \frown u$ of p with v , and the same first disagreement proves the same inequality. When $\alpha = 0$, the block V is empty. This covers all cases. $\square$

4.2 The exact predecessor identity

The following identity is the central combinatorial statement. It makes the exception at the old minimum completely explicit.

Theorem 4.4 (Predecessor and right-ancestor transport). *For every $x \in C_\alpha$ with $x > p$,*

$$\text{Pred}(\Theta_\alpha(x)) = \Theta_\alpha[\text{Pred}(x) \cup \{p\}]. \quad (8)$$

Consequently,

$$\mathbf{R}_s(\Theta_\alpha(x)) = \Theta_\alpha[\mathbf{R}_s(x)], \quad (9)$$

$$\mathbf{L}_s(\Theta_\alpha(x)) = \Theta_\alpha[\mathbf{L}_s(x) \cup \{p\}]. \quad (10)$$

At $x = p$ the image has no proper ancestors, while

$$\mathbf{R}_s(p) = \text{Pred}(p) = \{-\beta : \beta < \alpha\}, \quad \Theta_\alpha[\mathbf{R}_s(p)] = \{q_\beta : \beta < \alpha\}. \quad (11)$$

For $x, y > p$, simplicity is preserved and reflected:

$$x \prec y \iff \Theta_\alpha(x) \prec \Theta_\alpha(y).$$

Proof. First suppose $p \not\preccurlyeq x$. No prefix $v \preccurlyeq x$ can equal or extend p ; otherwise $p \preccurlyeq x$. Thus $\Theta_\alpha(v) = \langle + \rangle \frown v$ for all proper prefixes v of x . The proper prefixes of $\langle + \rangle \frown x$ are exactly the empty word, together with these $\langle + \rangle \frown v$. The empty word is $\Theta_\alpha(p)$, proving (8) in this case.

Now suppose $x = p \frown \langle + \rangle \frown u$. The proper prefixes of x are

$$\{-\beta : \beta \leq \alpha\} \cup \{p \frown \langle + \rangle \frown v : v \prec u\}.$$

Under Θ_α these become, respectively,

$$\{q_\beta : \beta < \alpha\} \cup \{0\}, \quad \{\langle + \rangle \frown p \frown v : v \prec u\}.$$

These are precisely the proper prefixes of $\langle + \rangle \frown p \frown u$. When u is empty the second collection is empty; when u has limit length it includes exactly its proper ordinal restrictions. This proves (8) without a missing limit case.

Filter (8) according to numerical comparison with $\Theta_\alpha(x)$. By lemma 4.3, comparison of images is comparison of preimages. Since $p < x$, the extra element $\Theta_\alpha(p) = 0$ is a left ancestor, never a right ancestor. This gives (9) and (10). Formula (11) follows directly from the all-minus word p . Finally, if $x, y > p$, membership of $\Theta_\alpha(x)$ in the predecessor set of $\Theta_\alpha(y)$ is, by injectivity and (8), exactly membership of x in $\text{Pred}(y)$. $\square$

Remark 4.5 (A stronger identity that would be false). For $\alpha > 0$ one must not assert (9) at $x = p$. Already $p = -1$ has right ancestor 0, whose image is 1, while $\Theta_\alpha(p) = 0$ has no ancestors at all. The missing relations at precisely this one point account for the extra condition in Theorem B.

Corollary 4.6 (Initiality of the image). *If Γ is initial and has minimum $p = -\alpha$, then $\Theta_\alpha[\Gamma]$ is initial and has minimum 0.*

Proof. The minimum assertion follows from the order-isomorphism. The image of p has no predecessors. For $x > p$ in Γ , the right side of (8) lies in $\Theta_\alpha[\Gamma]$, because both $\text{Pred}(x)$ and p lie in Γ . $\square$

Proposition 4.7 (The exact inverse obstruction). *Let $\Delta \subseteq \mathbf{No}_{\geq 0}$ be nonempty and initial. Put*

$$\Lambda_\alpha = \{0\} \cup \{q_\beta : \beta < \alpha\}.$$

Then $\Psi_\alpha[\Delta]$ is initial if and only if $\Lambda_\alpha \subseteq \Delta$.

Proof. The empty word 0 belongs to Δ , so p belongs to the inverse image. If the inverse image is initial, it contains every $-\beta$ with $\beta < \alpha$; applying Θ_α gives $q_\beta \in \Delta$.

Conversely, suppose $\Lambda_\alpha \subseteq \Delta$. All predecessors of p are then in the inverse image. For any other inverse-image element $x = \Psi_\alpha(y)$, where $y > 0$, (8) gives

$$\text{Pred}(x) \cup \{p\} = \Psi_\alpha[\text{Pred}(y)] \subseteq \Psi_\alpha[\Delta].$$

Hence every inverse-image element has all its predecessors in that image. $\square$

4.3 Exponent lengths and a finite picture

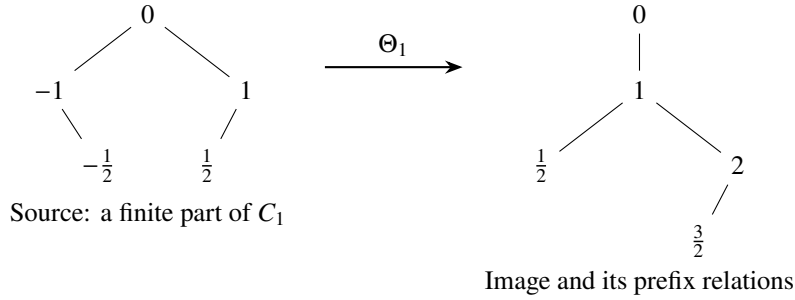
Proposition 4.8 (An exponent-birthday bound). *For every $x \in C_\alpha$,*

$$\ell(\Theta_\alpha(x)) \leq 1 +_{\text{ord}} \ell(x).$$

In particular, an infinite exponent length never increases.

Proof. At $x = p$ the image length is zero. In the outside case the length is exactly $1 + \ell(x)$. In the inside case, if $\ell(u) = \mu$, the source length is $\alpha + 1 + \mu$, and the image length is $1 + \alpha + \mu$. Monotonicity of ordinal addition gives $1 + \alpha + \mu \leq 1 + \alpha + 1 + \mu$. Finally, $1 + \lambda = \lambda$ for any infinite ordinal λ : write λ as a nonzero limit ordinal followed by a finite tail. $\square$

This is a bound for *exponents*. It is not a theorem bounding the birthday of the entire transformed surreal normal form.



Here $-1 \mapsto 0$, $0 \mapsto 1$, $1 \mapsto 2$, $-\frac{1}{2} \mapsto \frac{1}{2}$, and $\frac{1}{2} \mapsto \frac{3}{2}$. The numerical labels are exponents, not the monomials with those exponents.

5 Strong transport of normal forms

Definition 5.1 (Bottom-normalized transport). For $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$, $p = -\alpha$, and $c = 2^{-n}$, define

$$\mathcal{N}_{\alpha,n} \left(\sum_{\gamma \geq p} r_\gamma \omega^\gamma \right) = c^{-1} r_p + \sum_{\gamma > p} r_\gamma \omega^{\Theta_\alpha(\gamma)}. \quad (12)$$

A coefficient not in the support is interpreted as zero. The inverse is

$$\mathcal{N}_{\alpha,n}^{-1} \left(s_0 + \sum_{\delta > 0} s_\delta \omega^\delta \right) = c s_0 \omega^p + \sum_{\delta > 0} s_\delta \omega^{\Psi_\alpha(\delta)}. \quad (13)$$

Theorem 5.2 (Ambient transport). *The formulas above define inverse real-linear order-isomorphisms*

$$\mathcal{N}_{\alpha,n} : \mathcal{H}_{\mathbb{R}}(C_\alpha) \longrightarrow \mathcal{H}_{\mathbb{R}}(\mathbf{No}_{\geq 0}).$$

They preserve support cardinality and decreasing support order type, commute with every leading truncation, and preserve and reflect strong summability of set-indexed families. For every such strongly summable family,

$$\mathcal{N}_{\alpha,n} \left(\sum_{i \in I} x_i \right) = \sum_{i \in I} \mathcal{N}_{\alpha,n}(x_i). \quad (14)$$

For a nonzero x ,

$$\deg_H(\mathcal{N}_{\alpha,n}(x)) = \Theta_\alpha(\deg_H(x)), \quad \text{lc}(\mathcal{N}_{\alpha,n}(x)) = \begin{cases} c^{-1} \text{lc}(x), & \deg_H(x) = p, \\ \text{lc}(x), & \deg_H(x) > p. \end{cases} \quad (15)$$

Proof. An order-isomorphism takes a reverse-well-ordered support to a reverse-well-ordered support and preserves its order type. The support is still a set: it is the image of a set under a definable class function. Every coefficient is multiplied by a nonzero real number, so no support point is lost and no new point is created. These observations show that both displayed formulas are legitimate normal forms. Their inverse identities follow coefficient by coefficient from lemma 4.2.

Both addition of normal forms and the proposed map are coefficientwise, with the same fixed multiplier c^{-1} at the bottom point. Hence the map is real-linear. The greatest nonzero coefficient retains its sign, because all multipliers are positive. Thus the map and its inverse preserve order. This proves (15) as well.

A leading truncation is specified by an initial segment of the decreasing support enumeration. The same enumeration indices specify the corresponding truncation of the transported support. Applying the map therefore commutes with truncation, including at a limit index.

For a set-family (x_i) , one has the exact equality

$$\bigcup_i \text{supp}(\mathcal{N}_{\alpha,n}(x_i)) = \Theta_\alpha \left[\bigcup_i \text{supp}(x_i) \right].$$

Reverse well-ordering of these unions is equivalent in both directions. For each γ , the index set of family members contributing at γ is exactly the index set contributing at $\Theta_\alpha(\gamma)$ after transport. The point-finiteness requirement is therefore also equivalent. When it holds, coefficient extraction reduces (14) to a finite real sum, through which the fixed coefficient multiplier commutes. This proves the entire assertion without invoking convergence in an order topology. $\square$

Remark 5.3 (No implicit completion of a subgroup). The ambient Hahn spaces in theorem 5.2 are full spaces. A particular initial subgroup A need not be closed under all strong sums or all infinite subseries. We define its image as the actual set or class $\mathcal{N}_{\alpha,n}[A]$, not as the space of all series with the same coefficient groups. If a strongly summable family in A has sum outside A , (14) still holds in the ambient spaces, but is not an assertion of an internal summation operation on A .

Remark 5.4 (What is and is not preserved). The map preserves the ordered chain of Archimedean classes, with their exponents reindexed by Θ_α . It is not a homomorphism of additive *exponent* groups. Consequently (15) should not be described as preservation of a multiplicative valuation with the same value-group structure.

6 Proof of omnific normalization

We now prove Theorem A. Let A be as there, and obtain α, n, p, c from proposition 3.2. Write

$$\Gamma = \Gamma(A), \quad \Delta = \Theta_\alpha[\Gamma], \quad H = \mathcal{N}_{\alpha,n}[A].$$

The ambient transport restricts to an ordered-group isomorphism onto H . It remains to prove that this actual image is an initial subgroup of $\mathbf{Oz}$.

Lemma 6.1 (Transported coefficients). *The group H is truncation closed and cross-sectional on Δ , and its coefficient groups are*

$$H_0 = \mathbb{Z}, \quad H_{\Theta_\alpha(\gamma)} = A_\gamma \quad (\gamma > p). \quad (16)$$

Its exponent class is exactly Δ .

Proof. Truncation closure follows from theorem 5.2. For $\gamma > p$, one has $\mathcal{N}_{\alpha,n}(\omega^\gamma) = \omega^{\Theta_\alpha(\gamma)}$. At the bottom, $\mathcal{N}_{\alpha,n}(c\omega^p) = 1$. Thus all the leaders on Δ belong to H . No image element can have a support point outside Δ .

The formula at a positive image exponent leaves its coefficient unchanged, and the inverse formula has the same property. This proves both inclusions in the second identity of (16). At zero the coefficient group is $c^{-1}A_p = \mathbb{Z}$. $\square$

Proof of Theorem A. By corollary 4.6, Δ is initial with minimum zero. By lemma 6.1, the image is truncation closed and cross-sectional, and all its coefficient groups are initial real subgroups.

Consider an obligation in the dyadic right-ancestor condition for H : let $\delta, \eta \in \Delta$ and $\eta \in R_s(\delta)$. The point δ cannot be zero. Write $\delta = \Theta_\alpha(\gamma)$ with $\gamma > p$. By (9), there exists $\xi \in R_s(\gamma)$ with $\eta = \Theta_\alpha(\xi)$. Since $\xi > \gamma > p$, its coefficient group was not rescaled. The initiality of A gives

$$\mathbb{D} \subseteq A_\xi = H_\eta.$$

Every target obligation is therefore satisfied. The concrete sufficiency in [imported fact 2.2](#) proves that H is initial in **No**.

All exponents of H are nonnegative and its zero coefficient is an integer. Hence $H \subseteq \mathbf{Oz}$ by (4). Its minimum positive element is 1, either by order-isomorphism or directly from its normal forms. Finally, $N_{\alpha,n}(\varepsilon) = 1$, and all support and summability claims are the restrictions of [theorem 5.2](#). Uniqueness of α, n was established in [proposition 3.2](#). $\square$

Corollary 6.2 (The cited existence question). *Every discrete initial subgroup of **No** is isomorphic, as an ordered abelian group, to an initial subgroup of **Oz**.*

Remark 6.3 (An initial image is not a simplicity isomorphism). The conclusion is exactly an ordered-group isomorphism to an initial image. For example, on $A = \frac{1}{2}\mathbb{Z}$ the map $N_{0,1}$ sends 1 to 2 and $1/2$ to 1. Although $1 \prec 1/2$, one does not have $2 \prec 1$. Thus the normalization is not generally an isomorphism of the full simplicity-enriched structures. Any initial cyclic subgroup of **Oz** is $\mathbb{Z}$, and the order-isomorphism from $\frac{1}{2}\mathbb{Z}$ to $\mathbb{Z}$ is forced. Thus this example rules out a generally simplicity-preserving version. The published question does not demand that stronger conclusion.

7 Exact images, extremal groups, and admissible scales

7.1 The compulsory dyadic spine

Lemma 7.1 (The spine group is initial). *For every ordinal α , the group K_α of (1) is initial in **No**, has least positive element 1, and lies in **Oz**. Its exponent class is Λ_α .*

Proof. The proper prefixes of q_β are 0 and the q_η with $\eta < \beta$. Thus Λ_α is initial. The finite-support group on this class is cross-sectional and truncation closed. Its coefficient group is $\mathbb{Z}$ at zero and $\mathbb{D}$ at every other point. Any right ancestor lies among those positive spine points, where the required dyadic containment holds. Apply [imported fact 2.2](#). The remaining claims follow from its normal forms. $\square$

Proposition 7.2 (Necessary image data). *If A has least positive element $2^{-n}\omega^{-\alpha}$ and is initial, then $K_\alpha \subseteq N_{\alpha,n}[A]$.*

Proof. The image contains 1. For each $\beta < \alpha$, [corollary 3.3](#) gives $\mathbb{D}\omega^{-\beta} \subseteq A$. These terms are above the bottom exponent, so their coefficients are unchanged, and $\Theta_\alpha(-\beta) = q_\beta$. Thus the image contains every generating summand of K_α and hence all their finite sums. $\square$

Proof of Theorem B. Necessity has just been proved. Conversely, let $H \subseteq \mathbf{Oz}$ be initial with $K_\alpha \subseteq H$, and put $A = N_{\alpha,n}^{-1}[H]$. Since H contains 1, it has least positive element 1 and $H_0 = \mathbb{Z}$. Let $\Delta = \Gamma(H)$. The inclusion of K_α implies $\Lambda_\alpha \subseteq \Delta$, so $\Gamma = \Psi_\alpha[\Delta]$ is initial by [proposition 4.7](#). Its minimum is p .

By ambient transport, A is an additive, truncation-closed Hahn subgroup. Its bottom coefficient group is $c\mathbb{Z}$, which contains 1 since $c = 2^{-n}$; all other coefficient groups are inherited from H . It is therefore cross-sectional and has initial real coefficient groups.

We check the source right-ancestor condition. If $x \in \Gamma$ satisfies $x > p$ and $y \in R_s(x)$, then $\Theta_\alpha(y) \in R_s(\Theta_\alpha(x))$. The coefficient group at y is unchanged from that at $\Theta_\alpha(y)$, so it contains $\mathbb{D}$ by the initiality of H . If instead $x = p$, its right ancestors are exactly the $-\beta$ for $\beta < \alpha$. The hypothesis $K_\alpha \subseteq H$ says that all coefficient groups at their images q_β contain $\mathbb{D}$; again these coefficients are unchanged. This checks the only obligations not already covered by (9).

The criterion [imported fact 2.2](#) now proves A initial. Its least positive element is $N_{\alpha,n}^{-1}(1) = c\omega^p$, since the inverse is increasing. The inverse identities show uniqueness, and a fixed bijection preserves and reflects inclusion of its images. This proves the correspondence. $\square$

7.2 Smallest and largest initial realizations at a fixed scale

Define

$$L_{\alpha,n} = c\mathbb{Z}\omega^p \oplus \bigoplus_{\beta < \alpha} \mathbb{D}\omega^{-\beta}, \quad (17)$$

$$M_{\alpha,n} = \{x \in \mathcal{H}_{\mathbb{R}}(C_{\alpha}) : [\omega^p]x \in c\mathbb{Z}\}. \quad (18)$$

Here $[\omega^p]x$ denotes coefficient extraction. The direct sum in (17) again permits only finite supports; the class in (18) permits every set reverse-well-ordered support in the cone.

Theorem 7.3 (Extremal groups). *The groups $L_{\alpha,n}$ and $M_{\alpha,n}$ are initial with least positive element $c\omega^p$. Every initial group A with that least positive element satisfies*

$$L_{\alpha,n} \subseteq A \subseteq M_{\alpha,n}.$$

Moreover,

$$N_{\alpha,n}[L_{\alpha,n}] = K_{\alpha}, \quad N_{\alpha,n}[M_{\alpha,n}] = \mathbf{Oz}, \quad M_{\alpha,n} = c\omega^p \mathbf{Oz}. \quad (19)$$

Proof. The first two identities in (19) follow directly from the formulas: every positive image coefficient is unrestricted in M , whereas precisely the finite dyadic spine terms occur for L . By Theorem B, the inverse images of K_{α} and $\mathbf{Oz}$ are initial and have the required minimum. The lower bound on A follows from corollary 3.3 and $A_p = c\mathbb{Z}$; the upper bound follows from proposition 3.2.

For the last identity, ordinary multiplication by $c\omega^p$ sends a normal form on $[0, \infty)$ to one on $[p, \infty)$ by the arithmetic translation $\delta \mapsto p + \delta$. Its bottom coefficient belongs to $c\mathbb{Z}$, and the other coefficients range over all of $\mathbb{R}$ because multiplication by $c \neq 0$ is a bijection of $\mathbb{R}$. Conversely, translate any support in the cone back by $-p$ and divide its coefficients by c . The resulting series belongs to $\mathbf{Oz}$. $\square$

The equality $M_{\alpha,n} = \varepsilon \mathbf{Oz}$ does not make the theorem trivial. It concerns the *largest* group, whose higher coefficients are all real and whose support class is the full cone. Arithmetic translation can destroy initiality for smaller initial subgroups, even though it behaves correctly for this maximal one.

Corollary 7.4 (Sharp size obstruction). *Every set-sized initial group A with least positive element $2^{-n}\omega^{-\alpha}$ satisfies*

$$|A| \geq \max(\aleph_0, |\alpha|).$$

For every ordinal α and every $n \in \mathbb{N}$ this bound is attained by $L_{\alpha,n}$. In particular, a countable initial group can have such a least positive element only when α is countable.

Proof. The group contains the countably infinite cyclic subgroup generated by its least positive member and contains distinct terms $\omega^{-\beta}$ for $\beta < \alpha$. This gives the lower bound. The finite-support sum defining $L_{\alpha,n}$ has countable coefficient groups and an index set of size $|\alpha| + 1$, giving exactly the displayed cardinality. Its initiality and minimum follow from theorem 7.3. $\square$

This is a restriction on a specified initial realization. The ordinal α is not an invariant of the abstract ordered group under arbitrary isomorphisms: normalization itself changes the least positive element to 1.

7.3 Dyadic depth and the spectrum of inverse scales

Definition 7.5 (Dyadic-spine depth). Let $H \subseteq \mathbf{Oz}$ be a nonzero initial subgroup. If there exists an ordinal β such that $\mathbb{D}\omega^{q\beta} \not\subseteq H$, let $d(H)$ be the least such β . Otherwise write $d(H) = \mathbf{On}$ as a formal unbounded-depth marker, not as a surreal or a set ordinal.

Corollary 7.6 (Admissible scales). *If $d(H)$ is an ordinal, then for every $\alpha \in \mathbf{On}$ and every $n \in \mathbb{N}$,*

$$N_{\alpha,n}^{-1}[H] \text{ is initial} \iff \alpha \leq d(H).$$

If $d(H) = \mathbf{On}$, every ordinal α is allowed. Whenever the inverse is initial, its least positive member is $2^{-n}\omega^{-\alpha}$. The parameter n imposes no further obstruction.

Proof. By Theorem B, the condition is exactly that all $\mathbb{D}\omega^{q_\beta}$ for $\beta < \alpha$ lie in H . This holds precisely through the first missing index, inclusively, and fails at all larger indices. In the unbounded case there is no missing index. $\square$

For example, $d(K_\alpha) = \alpha$, since the exponent q_α is absent while all the earlier required dyadic terms are present. Thus the endpoint in corollary 7.6 can be a limit ordinal and must not be replaced with a strict inequality. For a set-sized H , a first missing index exists: the distinct monomials ω^{q_β} cannot all belong to a set. The subgroup $\bigcup_{\alpha \in \mathbf{On}} K_\alpha$, interpreted as the class of all finite dyadic-spine sums with an integer constant, is an example with unbounded depth. It is initial by imported fact 2.2, but is much smaller than $\mathbf{Oz}$, for example it does not contain ω^2 .

8 The quotient by the least positive cyclic subgroup

8.1 Convexity and an actual splitting

Fix A , p , c , and $\varepsilon = c\omega^p$ as before. Define

$$\pi_p : A \longrightarrow \mathbb{Z}, \quad \pi_p(x) = c^{-1}[\omega^p]x, \quad A^{>p} = \ker \pi_p.$$

Although its notation emphasizes the higher exponents, $A^{>p}$ also contains zero.

Lemma 8.1 (Bottom-layer decomposition). *The map π_p is a surjective additive homomorphism, $\pi_p(m\varepsilon) = m$, and*

$$A = A^{>p} \oplus \mathbb{Z}\varepsilon$$

as additive groups. The subgroup $\mathbb{Z}\varepsilon$ is convex. Every nonzero positive element of $A^{>p}$ dominates all integer multiples of ε .

Proof. Coefficient extraction is additive, and $A_p = c\mathbb{Z}$. This proves all assertions about π_p . Given $x \in A$, the term $\pi_p(x)\varepsilon$ belongs to A and subtraction produces an element of $A^{>p}$. The intersection of the two displayed summands is zero.

A nonzero element of $A^{>p}$ has greatest exponent strictly above p . Its absolute value therefore exceeds $m\varepsilon$ for every positive integer m . If $0 \leq x \leq m\varepsilon$ in A for some integer $m \geq 0$, its component in $A^{>p}$ must consequently be zero; thus $x \in \mathbb{Z}\varepsilon$. Translation and sign reversal prove convexity. $\square$

The summand $A^{>p}$ need not itself be initial as a concrete subclass of $\mathbf{No}$. For example, with $A = \mathbb{Z} + \mathbb{Z}\omega$, it is $\mathbb{Z}\omega$, which omits the proper prefix 1 of ω . A second exponent operation is needed for an *initial* quotient realization.

8.2 Deleting a positive root

Every positive surreal exponent has a unique form $\langle + \rangle \frown v$. Define

$$\rho : \mathbf{No}_{>0} \longrightarrow \mathbf{No}, \quad \rho(\langle + \rangle \frown v) = v.$$

Unlike Θ_α , this map has no exceptional point inside its domain. It preserves comparison and preserves and reflects prefix relations between positive source points.

Lemma 8.2 (Positive-root deletion). *If $\Delta \subseteq \mathbf{No}_{\geq 0}$ is initial, then $E = \rho[\Delta \setminus \{0\}]$ is initial, possibly empty. For $\delta > 0$,*

$$R_s(\rho(\delta)) = \rho[R_s(\delta)].$$

Proof. If $v \prec \rho(\delta)$, then $\langle + \rangle \frown v \prec \delta$ is a positive prefix and lies in Δ . Its image is v , establishing initiality. Every proper prefix of $\delta = \langle + \rangle \frown u$ is either 0, which is smaller than δ , or $\langle + \rangle \frown v$ with $v \prec u$. Comparisons of the latter with δ are comparisons of v with u . This proves the right-ancestor identity. $\square$

Let $H = \mathcal{N}_{\alpha,n}[A] \subseteq \mathbf{Oz}$ and

$$H^+ = \{h \in H : \text{ct}(h) = 0\}.$$

The superscript $+$ here means “strictly positive exponents,” not “positive elements.” To avoid any order-cone interpretation, define its transported group explicitly by

$$B = \left\{ \sum_{\delta > 0} s_\delta \omega^{\rho(\delta)} : \sum_{\delta > 0} s_\delta \omega^\delta \in H^+ \right\}. \quad (20)$$

Proposition 8.3 (An initial model of the quotient). *The class B in (20) is an initial additive subgroup of $\mathbf{No}$. The normal-form map in that formula induces ordered-group isomorphisms*

$$A/\mathbb{Z}\mathcal{E} \cong H/\mathbb{Z} \cong H^+ \cong B.$$

Proof. The first isomorphism is induced by $\mathcal{N}_{\alpha,n}$. Since constants of H are integers and all integers lie in H , every coset in $H/\mathbb{Z}$ has a unique representative with zero constant coefficient. The quotient ordering agrees with the ordering of those representatives, because a nonzero positive-exponent term dominates all integers.

The group H^+ is truncation closed and cross-sectional on $\Delta \setminus \{0\}$. Its coefficient groups at those points are the same as those of H . Reindexing by the order-isomorphism ρ produces a truncation-closed, cross-sectional subgroup with the same coefficient groups. Its exponent class is initial by lemma 8.2. The right-ancestor identity in that lemma transfers every dyadic obligation from H to B . Hence B is initial by imported fact 2.2. If there are no positive exponents, the group is $\{0\}$ and the same conclusion is immediate. $\square$

Proof of Theorem C. Combine lemma 8.1 and proposition 8.3. The explicit isomorphism sends $x \in A$ to

$$(\rho_*(\mathcal{N}_{\alpha,n}(x) - \pi_p(x)), \pi_p(x)),$$

where ρ_* denotes the normal-form reindexing of (20). The higher component determines the sign whenever it is nonzero; otherwise the sign is the sign of the integer component. This is precisely the indicated lexicographic order. $\square$

Corollary 8.4 (Finite iteration). *Starting with an initial group, repeatedly quotient by its least positive cyclic subgroup whenever the current quotient is nonzero and discrete. At every finite stage reached in this way, the quotient has an explicit initial realization in $\mathbf{No}$.*

Proof. Apply Theorem C at the first stage and then apply it to the initial realization just constructed at each subsequent stage. This is ordinary finite induction and makes no assertion at a limit stage. $\square$

Limit stages are treated in corollary 13.6, which rests on source 02: at every stage, limit stages included, the quotient has an explicit initial realization, it is the one above at finite stages, and the realizations at different stages are related by compression.

8.3 The converse construction: adjoining an integer bottom layer

There is also an explicit converse to the lexicographic decomposition. For an initial subgroup $B \subseteq \mathbf{No}$, including $B = \{0\}$, put

$$\sigma(\gamma) = \langle + \rangle \frown \gamma, \quad \sigma_* \left(\sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\sigma(\gamma)}, \quad J(B) = \mathbb{Z} \oplus \sigma_*[B].$$

The map σ_* changes exponents but not coefficients. Its image has no constant term, and each nonzero image element dominates every ordinary integer in absolute value.

Theorem 8.5 (Integer-bottom suspension). *The construction $B \mapsto J(B)$ gives an inclusion-preserving and inclusion-reflecting correspondence between initial subgroups of $\mathbf{No}$ and nonzero initial subgroups of $\mathbf{Oz}$. Its inverse sends H to the initial group obtained by deleting the leading plus sign from every positive exponent of $\ker(\text{ct}|_H)$. Moreover, $J(B) \cong B \overrightarrow{\times} \mathbb{Z}$ as ordered abelian groups.*

Proof. For nonzero B , let $E = \Gamma(B)$. The exponent class of $J(B)$ is $\{0\} \cup \sigma[E]$. It is initial: the prefixes of $\langle + \rangle \frown \gamma$ are zero and $\langle + \rangle \frown v$ for $v \prec \gamma$. The group is truncation closed and cross-sectional. Its coefficient group at zero is $\mathbb{Z}$, while the coefficient group at $\sigma(\gamma)$ is B_{γ} . Every right ancestor of $\sigma(\gamma)$ has the form $\sigma(v)$ with $v \in R_s(\gamma)$, so its coefficient group contains $\mathbb{D}$. The criterion [imported fact 2.2](#) proves initiality. For $B = \{0\}$, the construction is simply $J(B) = \mathbb{Z}$.

Conversely, if $H \subseteq \mathbf{Oz}$ is initial and nonzero, it contains $\mathbb{Z}$ and each constant coefficient of H is integral. Thus $H = \mathbb{Z} \oplus \ker(\text{ct}|_H)$. By [proposition 8.3](#), deleting the initial plus sign on the positive exponents of that kernel gives an initial group B . Reattaching the sign recovers the kernel exactly, including its actual infinite-support membership conditions. Hence $J(B) = H$, and the two constructions are inverse. Inclusions are preserved and reflected because both operations are fixed reindexings of the nonconstant normal forms. Finally, $(b, m) \mapsto \sigma_*(b) + m$ is additive and has the stated lexicographic order. $\square$

Corollary 8.6 (A relative criterion for discrete ordered groups). *Let G be a nonzero discrete ordered abelian group with least positive member e . Then G admits an initial realization in $\mathbf{No}$ if and only if $G/\mathbb{Z}e$ admits an initial realization and the exact sequence*

$$0 \longrightarrow \mathbb{Z}e \longrightarrow G \longrightarrow G/\mathbb{Z}e \longrightarrow 0$$

splits as a sequence of abelian groups.

Proof. If G admits an initial realization, the least positive element is carried to the least positive element there. Apply [Theorem C](#) and transport its splitting and quotient realization back. For any discrete ordered group, $\mathbb{Z}e$ is convex: finite induction shows that $[0, me]$ consists of $0, e, \dots, me$ for $m \in \mathbb{N}$. Conversely, a group-theoretic splitting identifies G with $(G/\mathbb{Z}e) \overrightarrow{\times} \mathbb{Z}$ as an ordered group: a lift of a nonzero positive quotient element lies above every element of $\mathbb{Z}e$, by convexity. Choose an initial realization B of the quotient and apply [theorem 8.5](#). $\square$

This criterion is relative: it reduces initial realizability to the quotient together with a splitting condition, rather than deciding initial realizability of every possible quotient.

No result of this section states that every convex quotient of every initial group has an initial realization; that was the extent of [source 01](#). The cyclic least-positive case is special here because it has a normal-form coefficient projection and the exact root-deletion operation above. [Source 02](#) removes the restriction: [Theorem D](#) realizes every convex factor, with the coefficient projection replaced by the difference of two leading truncations and root deletion replaced by sign-tree compression, and [proposition 13.5](#) shows that for $\mathbb{Z}\mathcal{E}$ its construction is exactly the one of this section.

9 Surcomplex and linear-algebraic consequences

9.1 Complex normal forms and rectangular modules

Let $\mathbf{No}[i] = \mathbf{No} \oplus i\mathbf{No}$ with $i^2 = -1$. Combining the real and imaginary supports gives a normal form with complex coefficients and surreal exponents; the union of the two reverse-well-ordered supports is again reverse-well-ordered. Conjugation acts coefficientwise. Define the Gaussian omnific module

$$\mathbf{Oz}[i] = \{a + ib : a, b \in \mathbf{Oz}\}.$$

For an additive group $A \subseteq \mathbf{No}$, put $A[i] = A \oplus iA$. It is a $\mathbb{Z}[i]$ -module because multiplication by i sends (a, b) to $(-b, a)$.

Definition 9.1 (Rectangular initiality). A rectangular subclass $A + iA$ is coordinatewise initial if A is initial in $\mathbf{No}$. This is a deliberately coordinate-dependent notion; no canonical single sign-tree order on $\mathbf{No}[i]$ is being postulated.

Theorem 9.2 (Surcomplex additive normalization). *Let A be a nonzero discrete initial subgroup and $H = \mathcal{N}_{\alpha,n}[A]$. The map*

$$\mathcal{N}_{\alpha,n}^{\mathbb{C}}(a + ib) = \mathcal{N}_{\alpha,n}(a) + i\mathcal{N}_{\alpha,n}(b)$$

is a $\mathbb{Z}[i]$ -linear isomorphism

$$A[i] \longrightarrow H[i] \subseteq \mathbf{Oz}[i].$$

It commutes with conjugation and has a complex-linear extension to the full complex-coefficient Hahn spaces on C_α and $\mathbf{No}_{\geq 0}$. That extension preserves and reflects set-indexed strong summability and commutes with its sums. The image is coordinatewise initial. For fixed α, n , its exact possible rectangular images are the $H[i]$ with $H \subseteq \mathbf{Oz}$ initial and $K_\alpha \subseteq H$.

Proof. Additivity and multiplication by i commute with the coordinatewise map, so it is $\mathbb{Z}[i]$ -linear. The inverse is coordinatewise $\mathcal{N}_{\alpha,n}^{-1}$. Conjugation negates the imaginary coordinate, and additivity gives the asserted compatibility.

In complex normal forms, the map is exactly

$$\sum_{\gamma \geq p} z_\gamma \omega^\gamma \longmapsto c^{-1} z_p + \sum_{\gamma > p} z_\gamma \omega^{\Theta_\alpha(\gamma)}.$$

The coefficient multiplier c^{-1} is real. It commutes with every complex scalar and with conjugation. The support argument in theorem 5.2 is unchanged; there is no additional infinite sum of constants because strong summability remains point-finite. Coordinatewise initiality follows from the initiality of H , and the range assertion is exactly Theorem B applied to the real factor. $\square$

9.2 What linear systems can be transported

Corollary 9.3 (Exact transport of integral linear equations). *Let $M \in \mathbb{Z}^{r \times s}$, let $b \in A^r$, and apply $\mathcal{N}_{\alpha,n}$ coordinatewise. Then*

$$Mx = b \iff M\mathcal{N}_{\alpha,n}(x) = \mathcal{N}_{\alpha,n}(b).$$

The corresponding statement holds for $M \in \mathbb{Z}[i]^{r \times s}$ and $b \in A[i]^r$ under $\mathcal{N}_{\alpha,n}^{\mathbb{C}}$. In the real case, coordinatewise inequalities are preserved as well.

Proof. Every row is a finite integer or Gaussian-integer linear combination. The relevant module isomorphism commutes with each such combination, and its injectivity gives the reverse implication. The real order assertion follows from the ordered-group isomorphism. $\square$

The right-hand side is transported too. In general $N_{\alpha,n}(1) \neq 1$, so this corollary must not be rewritten as preservation of equations with every ordinary constant fixed. Nor does it apply to nonlinear polynomials, arbitrary surreal scalar matrices, norms, or inner products.

10 Examples and failure modes

10.1 The motivating two-level example

Ehrlich and Kaplan's example in the paragraph preceding their question is

$$A = \mathbb{D} + \mathbb{Z}\omega^{-1}, \quad \varepsilon = \omega^{-1}.$$

Our construction has $\alpha = 1, n = 0, \Theta_1(-1) = 0$, and $\Theta_1(0) = 1$. It gives

$$d + m\omega^{-1} \mapsto d\omega + m, \quad H = \mathbb{D}\omega + \mathbb{Z} = K_1.$$

Thus the proposed general construction specializes to their known map, rather than treating that map as a new result [1, Section 9]. The example and the map stand in Section 9 of the journal version as well, immediately before its Question 9.1.

10.2 Rescale the bottom coefficient, not the whole series

Consider

$$A = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\omega.$$

The exponent class $\{0, 1\}$ is initial, there are no right-ancestor dyadic obligations, and its coefficient groups $\frac{1}{2}\mathbb{Z}$ and $\mathbb{Z}$ are initial. Hence A is initial by imported fact 2.2. Its least positive element is $1/2$.

Scalar multiplication by 2 yields $\mathbb{Z} + 2\mathbb{Z}\omega$, which omits ω despite containing 2ω . Equivalently, its coefficient group at exponent 1 is not initial in $\mathbb{R}$. By contrast,

$$N_{0,1}(a/2 + b\omega) = a + b\omega$$

has initial image $\mathbb{Z} + \mathbb{Z}\omega$. Only the bottom coefficient changes.

10.3 An exponent translation fails at the first infinite branch

Let

$$\Gamma = \{-1\} \cup \mathbb{N} \cup \{\omega\}$$

be an exponent class, and let A be the finite-support group with coefficient $\mathbb{Z}$ at -1 , coefficient $\mathbb{D}$ at 0, and coefficient $\mathbb{Z}$ at every positive member of Γ . This exponent class is initial: -1 contributes the prefix 0, and the ordinal ω contributes all finite ordinals. Its only right-ancestor obligation is at -1 , forcing the chosen dyadic group at zero. Thus A is initial, with least positive element ω^{-1} .

Multiplication by ω replaces the support class by

$$\Gamma + 1 = \{0, 1, 2, \dots\} \cup \{\omega + 1\}.$$

This class is not initial because it contains $\omega + 1$ but omits its prefix ω . The resulting group is not initial. In contrast, root relocation satisfies

$$\Theta_1(-1) = 0, \quad \Theta_1(k) = k + 1 \ (k < \omega), \quad \Theta_1(\omega) = \omega.$$

The last identity uses $\langle + \rangle \frown \langle + \rangle^\omega = \langle + \rangle^\omega$; it is precisely where ordinal concatenation differs from arithmetic translation. The correct image has exponent class $\mathbb{N} \cup \{\omega\}$, coefficient $\mathbb{Z}$ at zero, $\mathbb{D}$ at one, and $\mathbb{Z}$ at the other exponents. It is initial.

10.4 A genuine limit-support example

Take $\alpha = \omega$, $n = 0$, and consider the group of all series

$$x = \sum_{k < \omega} a_k \omega^{-k} + m \omega^{-\omega}, \quad a_k \in \mathbb{D}, \quad m \in \mathbb{Z}. \quad (21)$$

Zeros in this expression are ignored. The allowed support is a subset of the reverse-well-ordered sequence $0, -1, -2, \dots, -\omega$. The exponent class is initial, its ancestor coefficients are dyadic, and its bottom coefficient is integral, so the full group (21) is initial.

Since $\Theta_\omega(-k) = q_k = 2^{-k}$ for finite k and $\Theta_\omega(-\omega) = 0$, normalization gives the exact strong-sum identity

$$\mathcal{N}_{\omega,0}(x) = \sum_{k < \omega} a_k \omega^{2^{-k}} + m. \quad (22)$$

For example, when $a_k = 1$ for every k and $m = 1$, the source and image have support order type $\omega + 1$. The decreasing exponents $1, 1/2, 1/4, \dots, 0$ are a legitimate reverse-well-ordered support; no analytic limiting process is involved.

The finite-support minimal group $L_{\omega,0}$ is a proper subgroup of the group in (21). Its image K_ω consists only of finite sums of the positive terms appearing in (22), together with an integer constant. The full limit-support example therefore illustrates why the image cannot be reconstructed from its coefficient groups alone.

10.5 Inverse initiality needs the whole dyadic spine

Let $H = \mathbb{Z}$ and $\alpha = 1, n = 0$. The inverse formula produces

$$\mathcal{N}_{1,0}^{-1}[\mathbb{Z}] = \mathbb{Z}\omega^{-1},$$

which is not initial: 1 is simpler than ω^{-1} but is missing. Here $K_1 \not\subseteq H$, so Theorem B correctly rules out initiality.

There is also a purely coefficient-level obstruction. Let $H = \mathbb{Z} + \mathbb{Z}\omega$ and keep $\alpha = 1, n = 0$. Its exponent class contains $\Lambda_1 = \{0, 1\}$, so the inverse exponent class is initial. Nevertheless, the inverse group is

$$\mathbb{Z} + \mathbb{Z}\omega^{-1},$$

which is not initial: its monomial ω^{-1} forces every dyadic constant through (3), but its constant group is only $\mathbb{Z}$. Equivalently, $1/2$ is a missing prefix of ω^{-1} . Thus the exponent condition in proposition 4.7 alone does not suffice for the group-level inverse.

10.6 Multiplication and ordinary constants are not preserved

For $A = \mathbb{D} + \mathbb{Z}\omega^{-1}$, one has $1 \in A$ and $\mathcal{N}_{1,0}(1) = \omega$. The product $1 \cdot 1$ also belongs to A , but

$$\mathcal{N}_{1,0}(1 \cdot 1) = \omega \neq \omega^2 = \mathcal{N}_{1,0}(1)\mathcal{N}_{1,0}(1).$$

Even on products that happen to stay in the source, multiplicativity fails. The surcomplex extension inherits this limitation. In particular, no preservation of the complex modulus or of multiplication by general surcomplex scalars follows from its Gaussian-linearity.

11 Convex subgroups as exponent cuts

This section and the next seven are source 02's (section 1.5). They extend section 8 from the least positive cyclic subgroup to every convex subgroup.

11.1 Conventions of the second source

All statements about arbitrary set-sized ordered groups can be read in ZFC. For the uniform class constructions we work, as in section 2, in NBG with global choice, allowing class parameters. A quotient of a proper-class group is not treated as a set of proper-class cosets. When such notation is used, the quotient is represented by the explicitly constructed normal-form complement. The class operations restrict to set operations on every individual support. In particular, none of the proofs sums a proper class of terms.

Signs, prefixes, Pred , R_s , normal forms, leading truncations, strong summability, G_γ and cross-sectionality are as in section 2. For two sign sequences x, y , let $x \wedge y$ be their longest common prefix; if one is a prefix of the other, their meet is the shorter one. Source 02 uses the Ehrlich–Kaplan criterion in exactly the concrete form of imported fact 2.2, including its proof that the canonical normal-form image is initial, and the description of initial real coefficient groups in imported fact 2.3; on an active exponent such a group is nonzero and contains 1. It imports one more result.

Imported fact 11.1 (Ehrlich’s divisible embedding). *Every divisible ordered abelian group has an initial realization in **No** [7].*

This is Ehrlich’s earlier embedding theorem, also recalled in [1]. It is used below only for set-sized groups. Neither source re-proves imported fact 2.2 or imported fact 11.1.

Remark 11.2 (Actual membership is part of the data). A truncation-closed subgroup need not contain every series allowed by its individual coefficient groups. In particular it need not be closed under arbitrary infinite subseries or all ambient strong sums. All transported groups below are actual images, not silently enlarged Hahn completions. This is the caution of remark 5.3, stated by both sources.

Several symbols of source 02 are renamed here, because the letters are already in use in sections 1 to 10. The renaming changes no statement.

here	in source 02	reason for the change
res_G	ρ_G	ρ is root deletion (lemma 8.2)
G^{div}	$D(G)$	$\mathbb{D}$ is $\mathbb{Z}[1/2]$
$\zeta, \zeta_* \in \widehat{\mathbb{Z}}; G_\zeta$	$\alpha, \alpha_*; G_\alpha$	α is the ordinal of the bottom exponent $-\alpha$
τ_*	p_*	$p = -\alpha$ is the bottom exponent
$W = \mathbb{Q}\omega^2 + \mathbb{Z}\omega + \mathbb{Q}$	K	K_α is the spine group
$\widetilde{G}_\zeta = G_\zeta \overrightarrow{\times} \mathbb{Q}$	H_α	H is the image in Theorem A
$C \subseteq C'$	$C_1 \subseteq C_2$	C_α is the cone $[-\alpha, \infty)$
$\Gamma_C, \Gamma^C = \Gamma \setminus \Gamma_C$	Λ_C, Δ	Λ_α is the spine exponent class
$C^\natural = G[\Gamma^C]$	H_C	as for H
b_E, ι_x	c_E, e_x	$c = 2^{-n}$; e is a least positive element
π_1	π	π_p is the bottom coefficient projection
Theorems D, E, F	Theorems A, B, C	source 01’s letters are kept

Kept unchanged: e for the least positive element of an abstract discrete group (as in corollary 8.6); Π for the purely infinite normal forms, zero included (as in section 1.4); $J(B)$ for the suspension of theorem 8.5; “coordinatewise initial” of definition 9.1, which source 02 calls “rectangularly initial”. In sections 14 to 17 the letter n is an ordinary modulus, not the dyadic exponent of proposition 3.2. The profinite residue res_G is the residue res_I of the trigonometry report (`trigonometry:per:eq:residue`), and G^{div} its I^{div} (`trigonometry:per:prop:divisible`).

11.2 Exponent cuts and the canonical splitting

Throughout sections 11 to 13 let $G \subseteq \mathbf{No}$ be initial, with active exponent class $\Gamma = \Gamma(G)$. For an exponent subclass E define the ambient coefficient restriction

$$P_E(x) = \sum_{\gamma \in \text{supp}(x) \cap E} r_\gamma \omega^\gamma.$$

It is always an ambient normal form. Its membership in G needs proof.

Lemma 11.3 (Archimedean detection). *Let C be a convex subgroup of G and set*

$$\Gamma_C = \{\gamma \in \Gamma : \omega^\gamma \in C\}.$$

Then Γ_C is a numerical lower segment of Γ . For nonzero $x \in G$, with greatest exponent γ , the following are equivalent:

$$x \in C; \quad \gamma \in \Gamma_C; \quad \text{supp}(x) \subseteq \Gamma_C.$$

Proof. The numbers $|x|$ and ω^γ are Archimedean equivalent: each is bounded by a positive ordinary integer multiple of the other. Indeed their ratio has positive, nonzero real leading coefficient and an infinitesimal remainder. Convexity and subgroup closure therefore show that $x \in C$ exactly when $\omega^\gamma \in C$. If $\delta < \gamma$ lies in Γ , then $0 < \omega^\delta < \omega^\gamma$, so membership of the latter in C forces membership of the former. This proves the lower-segment assertion and the support equivalence. Zero belongs to every subgroup separately. $\square$

Lemma 11.4 (Restriction to a convex interval). *If E is a convex subclass of the ordered class Γ , then $P_E(G) \subseteq G$. More precisely, P_E is the difference of two leading-truncation projections. Its image equals*

$$G[E] := \{x \in G : \text{supp}(x) \subseteq E\}.$$

This subgroup is truncation closed and cross-sectional on E .

Proof. There is nothing to prove when E is empty. Otherwise set

$$U = \{\gamma \in \Gamma : (\exists \delta \in E) \delta \leq \gamma\}, \quad V = \{\gamma \in \Gamma : (\forall \delta \in E) \delta < \gamma\}.$$

Both are upper segments, $V \subseteq U$, and convexity gives $E = U \setminus V$. For every reverse-well-ordered support, restriction to an upper segment retains a leading segment. Truncation closure, with zero and the entire form admitted, gives $P_U(x), P_V(x) \in G$. Hence $P_E(x) = P_U(x) - P_V(x) \in G$. The image description follows from $P_E^2 = P_E$. Leading truncations of an element supported in E remain supported there, and every ω^γ with $\gamma \in E$ lies in the image. $\square$

Theorem 11.5 (Canonical splitting in a chosen realization). *For a convex subgroup C of G , put $\Gamma^C = \Gamma \setminus \Gamma_C$. Then*

$$C = G[\Gamma_C], \quad C^\natural := G[\Gamma^C], \quad G = C^\natural \oplus C.$$

The map $x \mapsto P_{\Gamma^C}(x)$ realizes G/C as the ordered subgroup $C^\natural$, and addition induces an ordered-group isomorphism

$$C^\natural \overrightarrow{\times} C \longrightarrow G, \quad (h, c) \longmapsto h + c.$$

Proof. The first equality is lemma 11.3. Since Γ^C is an upper segment, $P_{\Gamma^C}(x)$ is a leading truncation in G . The difference $x - P_{\Gamma^C}(x)$ is supported in Γ_C and belongs to C . The supports of the two summands are disjoint, proving uniqueness. If $h \neq 0$ and $c \in C$, the leading exponent of h exceeds every exponent of c . Consequently $h + c$ has the sign of h ; when $h = 0$ it has the sign of c . This is the lexicographic ordering. The kernel of P_{Γ^C} is C , so it gives the stated quotient model and the inclusion of $C^\natural$ is an order-preserving section. $\square$

For a discrete initial A and $C = \mathbb{Z}\varepsilon$ one has $\Gamma_C = \{p\}$ and $C^\natural = A^{>p}$, and theorem 11.5 is lemma 8.1 (see proposition 13.5).

Remark 11.6 (Which projection preserves order?). The upper projection P_{Γ_C} is nondecreasing. The lower projection P_{Γ_C} usually is not: for example, a positive element $\omega - 1$ has negative constant projection. Splitting the abelian-group sequence does not make both component projections positive maps.

Proposition 11.7 (Convex factors and commuting projections). *If $C \subseteq C'$ are convex in G , then $\Gamma_C \subseteq \Gamma_{C'}$ and $E = \Gamma_{C'} \setminus \Gamma_C$ is convex in Γ . Restriction induces an order-isomorphism*

$$C'/C \cong G[E].$$

For arbitrary exponent subclasses E, F the ambient identities are

$$P_E P_F = P_{E \cap F} = P_F P_E. \quad (23)$$

If E, F are convex intervals, all these maps preserve G .

Proof. The lower-segment descriptions give the interval assertion. Every $x \in C'$ splits uniquely into its E -part and its Γ_C -part. The first is a representative of the quotient, and its leading coefficient determines the sign of a nonzero coset. The identities (23) are coefficientwise. Intersections of convex intervals are convex, so lemma 11.4 handles membership. $\square$

These arguments already yield an intrinsic obstruction: an ordered abelian group with a nonsplit convex subgroup sequence cannot be initially realizable. The next step shows that the explicit complements and interval subgroups can themselves be given initial realizations.

12 Compression of a meet-closed sign tree

The central construction is elementary at the level of signs, but its retained positions must be indexed by their *order type*, not by ordinary integer subtraction.

Definition 12.1 (Retained-ancestor compression). A nonempty subclass $E \subseteq \mathbf{No}$ is *meet closed* if $x \wedge y \in E$ for all $x, y \in E$. For $x \in E$ let

$$I_E(x) = \{\beta < \ell(x) : x \upharpoonright \beta \in E\}, \quad \kappa_E(x) = \text{otp}(I_E(x)).$$

Let $\iota_x : \kappa_E(x) \rightarrow I_E(x)$ be the unique increasing enumeration. Define $b_E(x)$ by

$$\ell(b_E(x)) = \kappa_E(x), \quad b_E(x)(\xi) = x(\iota_x(\xi)). \quad (24)$$

Thus the sign at position β is retained exactly when its preceding prefix belongs to E .

The set $I_E(x)$ is a subset of the ordinal $\ell(x)$, even when E is a proper class. The definition is therefore elementwise legitimate.

Lemma 12.2 (Convex intervals are meet closed). *If Γ is initial in $\mathbf{No}$ and $E \subseteq \Gamma$ is a nonempty numerically convex subclass, then E is meet closed.*

Proof. If one of $x, y \in E$ is a prefix of the other, their meet is an endpoint. Otherwise, their common prefix $m = x \wedge y$ belongs to Γ by initiality. At the first differing sign, one continuation is minus and the other plus. Thus m lies strictly between x and y numerically, and convexity puts m in E . $\square$

Lemma 12.3 (Retained prefix formula). *If $y = x \upharpoonright \beta \in E$ with $\beta < \ell(x)$, then*

$$I_E(y) = I_E(x) \cap \beta.$$

Writing $\xi = \text{otp}(I_E(x) \cap \beta)$, one has

$$b_E(y) = b_E(x) \upharpoonright \xi, \quad \xi < \kappa_E(x), \quad \iota_x(\xi) = \beta.$$

Proof. The prefixes of y are exactly the prefixes of x of length less than β . Since $\beta \in I_E(x)$, it is the next retained position after those earlier positions. The signs in the two compressed sequences agree at all earlier retained positions by definition. $\square$

Theorem 12.4 (Compression theorem). *For nonempty meet-closed $E \subseteq \mathbf{No}$, the map $b_E : E \rightarrow \mathbf{No}$ is an order-isomorphism onto an initial subclass $E^b := b_E[E]$. For $x, y \in E$,*

$$y < x \iff b_E(y) < b_E(x), \quad (25)$$

$$b_E(x \wedge y) = b_E(x) \wedge b_E(y), \quad (26)$$

$$R_s(b_E(x)) = b_E[R_s(x) \cap E]. \quad (27)$$

It also preserves and reflects left-ancestor relations.

Proof. First suppose $y < x$. The retained prefix formula shows that $b_E(y)$ is a proper prefix of $b_E(x)$. At its first new position the compressed continuation has sign $x(\ell(y))$. Consequently the numerical comparison between y and x is unchanged.

Now suppose x, y are incomparable by prefix and put $m = x \wedge y$. Meet closure gives $m \in E$. The retained positions preceding $\ell(m)$ are identical for x and y , and the position $\ell(m)$ itself is retained in both. Their compressed sequences therefore have common prefix $b_E(m)$ and then opposite signs. The first differing sign has not changed. This proves order preservation, injectivity, and (26) in the incomparable case. The comparable case of the meet identity follows from the retained prefix formula. These alternatives also prove the reverse implication in (25): incomparable points remain incomparable.

Let z be an arbitrary proper prefix of $b_E(x)$, of length $\xi < \kappa_E(x)$. Put $\beta = \iota_x(\xi)$ and $y = x \upharpoonright \beta$. By definition $y \in E$, and the retained prefix formula gives $b_E(y) = z$. Thus E^b is initial, at successor and limit prefix lengths alike. Combining this description of every prefix with order preservation proves (27); the left-ancestor assertion is identical with the inequality reversed. $\square$

Remark 12.5 (No unjustified limit step). A limit ordinal $\xi < \kappa_E(x)$ still has the position $\iota_x(\xi)$. The proof uses that position directly. It does not assume E is closed under limits of chains or that an infinite sequence of deletions converges to a surreal. This is why the construction applies to cuts without endpoints.

Proposition 12.6 (Uniqueness and nested coherence). *The map b_E is the unique isomorphism of the numerical and prefix orders from E onto an initial subclass of $\mathbf{No}$. If $F \subseteq E$ is nonempty and meet closed, then*

$$b_{b_E[F]}(b_E(x)) = b_F(x) \quad (x \in F). \quad (28)$$

Proof. In an initial image, the proper prefixes of the image of x correspond exactly to the proper ancestors of x in E , in their simplicity order. Hence the image length must be $\kappa_E(x)$. The sign at the position corresponding to $y < x$ is plus exactly when $y < x$. Both the positions and signs are forced; they agree with (24).

By (26), the subclass $b_E[F]$ is meet closed. In a second compression, a sign of $b_E(x)$ is retained exactly when its preceding prefix is $b_E(y)$ for some $y \in F$. By the retained prefix formula and prefix reflection, these are exactly the original positions β with $x \upharpoonright \beta \in F$. Thus the two-stage operation retains the same well-ordered positions, in the same order, as direct compression by F . This proves (28). $\square$

Example 12.7 (Compression is not exponent translation). Take the full finite sign tree of lengths at most two, in numerical order:

$$-- < - < -+ < 0 < +- < + < ++.$$

On the convex interval $E = \{-+, 0, +-\}$, compression gives

$$-+ \mapsto -, \quad 0 \mapsto 0, \quad +- \mapsto +.$$

The original exponents are $-1/2, 0, 1/2$, and their images are $-1, 0, 1$. On $E = \{+, ++\}$ the images are $0, +$. The definition is a tree operation, not a uniform rule of subtracting a real or surreal endpoint.

Example 12.8 (Meet closure cannot be omitted). For $E = \{-, +\}$ neither point has a proper prefix in E . Both therefore compress to 0, so injectivity fails. This set omits the meet 0. Convexity in an initial exponent class prevents precisely this defect.

The compression theorem is compatible with the general viewpoint of surreal substructures and convex partitions in [5]. It does not assert that a set-sized convex interval is a copy of the whole proper-class structure $\mathbf{No}$; its target is an arbitrary initial subclass, often small.

13 Initial realizations of all convex factors

For the empty interval we set $E^b = \emptyset$, $\mathcal{F}_E(G) = \{0\}$, and let Φ_E be the unique map between the zero Hahn spaces. This convention also covers equal convex subgroups in the factor statements.

Theorem 13.1 (Hahn transport on an interval). *Let $G \subseteq \mathbf{No}$ be initial with exponent class Γ , and let $E \subseteq \Gamma$ be a nonempty convex interval. Define*

$$\Phi_E \left(\sum_{\gamma \in E} r_\gamma \omega^\gamma \right) = \sum_{\gamma \in E} r_\gamma \omega^{b_E(\gamma)}. \quad (29)$$

Then Φ_E is an ordered real-linear isomorphism $\mathcal{H}_{\mathbb{R}}(E) \rightarrow \mathcal{H}_{\mathbb{R}}(E^b)$. It restricts to an ordered-group isomorphism of $G[E]$ onto an initial subgroup $\mathcal{F}_E(G)$ of $\mathbf{No}$. It preserves support order types, leading coefficients, leading truncations, and strong summability in both directions.

Proof. By lemma 12.2 and theorem 12.4, the exponent map is an order-isomorphism with initial image. It therefore sends each set-sized reverse-well-ordered support to another such support, with the same order type. Its inverse has the same property. Coefficients are unchanged, so addition and real scalar multiplication commute with the map, and the leading coefficient retains its sign.

Set $B = \Phi_E[G[E]]$. By lemma 11.4, the source subgroup is cross-sectional and truncation closed. These properties pass to B , and its coefficient group at $b_E(\gamma)$ is exactly G_γ . The latter is initial in $\mathbb{R}$. Every right ancestor of $b_E(\gamma)$ has the form $b_E(\delta)$ with $\delta \in R_s(\gamma) \cap E$, by (27). The original dyadic obligation (3) supplies the required coefficient group. The sufficiency direction of imported fact 2.2 now shows that the actual image B is initial.

Leading truncations retain the same initial sets of support indices before and after reindexing. For a set-family (x_i) , the union of its transported supports is the image under b_E of the original union. Reverse well-ordering is equivalent in both directions. At any exponent, the set of indices contributing a nonzero coefficient is unchanged. Point-finiteness is therefore also equivalent. In a strongly summable family, every resulting coefficient is a finite real sum, and the transport commutes with that sum. $\square$

This is the analogue, for an interval and without a rescaled bottom coefficient, of theorem 5.2; the proof has the same structure.

Proof of Theorem D. Transport G to an actual initial realization. Apply proposition 11.7 to obtain $C'/C \cong G[\Gamma_{C'} \setminus \Gamma_C]$. If the exponent interval is empty, the factor is zero. Otherwise theorem 13.1 gives the required initial image. Splitting is theorem 11.5; the composition laws are (23) and (28). $\square$

Corollary 13.2 (Subgroups, quotients, and transfinite cuts). *The class Init is closed under convex subgroups, convex ordered quotients, and their subquotients. For a chain of convex subgroups $(C_\xi)_{\xi < \lambda}$ of an initially realizable group, any union or intersection which is a convex subgroup has the same conclusion for itself and for the corresponding quotient. No successor-stage restriction is necessary.*

Proof. The first assertions are special cases of Theorem D. A union of an increasing chain and an intersection of convex subgroups are convex. The theorem applies directly to their exponent cuts, rather than asserting that images previously chosen at earlier stages have a particular union or intersection. $\square$

Corollary 13.3 (Coherent compressed factors). *For nonempty convex intervals $F \subseteq E \subseteq \Gamma$, the direct map Φ_F agrees with first transporting by Φ_E and then compressing the interval $\flat_E[F]$. Thus the two constructions give the same normal forms, not just abstractly isomorphic groups.*

Proof. The coefficient at every retained exponent is unchanged in both constructions, and (28) identifies the final exponents. Actual membership follows from the restriction and image definitions. $\square$

Remark 13.4 (Strong sums and factor projection). The reindexing Φ_E preserves and reflects strong summability on its ambient space. The restriction P_E preserves strong summability but need not reflect it, because terms outside E are discarded. Neither assertion enlarges G to a group closed under all ambient strong sums.

13.1 Examples with an infinite exponent cut

Let $G = \mathbb{Q}[\omega]$ additively, meaning finite sums $\sum_{j=0}^m q_j \omega^j$. The exponent class $\mathbb{N}$ consists of all finite all-plus sequences and is initial. There are no right-ancestor obligations within this class, so the criterion shows that G is initial. The convex subgroup of degree less than k has quotient represented by degrees at least k ; compression sends the ordinal exponent $k + j$ to j . This recovers an initial quotient isomorphic to G .

For a genuinely transfinite example, take finite-support rational forms on the initial exponent class of all ordinals below $\omega + \omega$. The lower segment of finite exponents defines a convex subgroup. On the upper interval $[\omega, \omega + \omega)$, compression deletes the first ω ancestor positions and sends $\omega + n$ to n . The quotient again has an initial rational-polynomial realization. This calculation uses sign sequences, not an assumption that general surreal subtraction preserves initiality.

13.2 The cyclic case is root relocation followed by root deletion

The following statement, added in the merge, identifies source 02's construction for $C = \mathbb{Z}\varepsilon$ with the construction of section 8. Neither source states it.

Proposition 13.5 (Compression of the bottom point; [merge]). *Let A be a nonzero discrete initial subgroup, with $\alpha, n, p = -\alpha, c, \varepsilon = c\omega^p, \Theta_\alpha, N_{\alpha,n}, \rho, \pi_p, A^{>p}, H$ and B as in sections 3 to 6 and 8, and let $\Gamma = \Gamma(A)$ and $C = \mathbb{Z}\varepsilon$. Then $\Gamma_C = \{p\}$, $E := \Gamma^C = \Gamma \setminus \{p\}$, $C^\natural = A[E] = A^{>p}$, and*

$$\flat_E(\gamma) = \rho(\Theta_\alpha(\gamma)) \quad (\gamma \in E), \quad \text{that is,} \quad \flat_E(p \frown \langle + \rangle \frown u) = p \frown u, \quad \flat_E(x) = x \quad (p \not\leq x). \quad (30)$$

Consequently $\Phi_E(P_E(x)) = \rho_(N_{\alpha,n}(x) - \pi_p(x))$ for every $x \in A$, the realization $\mathcal{F}_E(A)$ of $A/\mathbb{Z}\varepsilon$ is the group B of (20), and the splitting of theorem 11.5 for $C = \mathbb{Z}\varepsilon$ is that of lemma 8.1 and Theorem C.*

Proof. Since $\omega^p = 2^n \varepsilon \in \mathbb{Z}\varepsilon$ and no other monomial ω^γ , $\gamma \in \Gamma$, lies in the convex subgroup $\mathbb{Z}\varepsilon$ (its positive members are the multiples of $c\omega^p$), $\Gamma_C = \{p\}$. By lemma 11.3, an element of A lies in $A[E]$ exactly when its support avoids p , that is, when $\pi_p(x) = 0$; so $A[E] = A^{>p}$.

First route: the formula. By proposition 3.2 every $x \in E$ lies in the cone C_α and $x > p$, so, as in section 4, either $p \not\leq x$ or $x = p \frown \langle + \rangle \frown u$. Every proper prefix of x lies in the initial class Γ , hence in E unless it equals p . If $p \not\leq x$, no prefix equals p , every position is retained and $b_E(x) = x = \rho(\langle + \rangle \frown x) = \rho(\Theta_\alpha(x))$. If $x = p \frown \langle + \rangle \frown u$, the only prefix equal to p is $x \upharpoonright \alpha$, so exactly the plus sign at position α is deleted and $b_E(x) = p \frown u = \rho(\langle + \rangle \frown p \frown u) = \rho(\Theta_\alpha(x))$. Positions are read as ordinal intervals, so limit lengths of u need no separate case.

Second route: uniqueness. On E , the map $\rho \circ \Theta_\alpha$ is an order-isomorphism (lemma 4.3, and ρ preserves comparison), it preserves and reflects prefixes (theorem 4.4 for points above p , and lemma 8.2), and its image $\rho[\Theta_\alpha[\Gamma] \setminus \{0\}]$ is initial (corollary 4.6 and lemma 8.2). By the uniqueness in proposition 12.6 it equals b_E .

For $x = \sum_\gamma r_\gamma \omega^\gamma \in A$, (12) gives $N_{\alpha,n}(x) - \pi_p(x) = \sum_{\gamma > p} r_\gamma \omega^{\Theta_\alpha(\gamma)}$, since the constant term of $N_{\alpha,n}(x)$ is $c^{-1}r_p = \pi_p(x)$. Deleting the root of each exponent gives $\sum_{\gamma \in E} r_\gamma \omega^{\rho \Theta_\alpha(\gamma)}$, which is $\Phi_E(P_E(x))$ by (30). An element of H has zero constant coefficient exactly when it is the image of an element of $A^{>p}$, so $H^+ = N_{\alpha,n}[A^{>p}]$ and $B = \rho_*[H^+] = \Phi_E[A[E]] = \mathcal{F}_E(A)$. The two splittings have the same complement $A^{>p} = C^\natural$ and the same cyclic summand. $\square$

Thus the construction of source 02, applied to $C = \mathbb{Z}\varepsilon$, compresses away the bottom exponent and gives exactly source 01's $\rho \circ \Theta_\alpha$ on exponents, with the same coefficients. The two constructions also agree by the uniqueness of compression, and a finite computation at placement confirmed (30) on words of small length (section 19.5).

13.3 Limit stages of iterated bottom-layer removal

Corollary 8.4 (source 01) covers finite stages only. The following corollary, formulated in the merge from source 02's corollaries 13.2 and 13.3 and proposition 13.5, covers every stage.

Corollary 13.6 (Transfinite iteration; [merge]). *Let $A \subseteq \mathbf{No}$ be initial with exponent class Γ . Define subgroups C_ξ of A by recursion: $C_0 = \{0\}$; if A/C_ξ is nonzero and discrete, $C_{\xi+1}$ is the preimage in A of the least positive cyclic subgroup of A/C_ξ , and otherwise the recursion stops at ξ ; at a limit λ , $C_\lambda = \bigcup_{\xi < \lambda} C_\xi$. Put $E_\xi = \Gamma \setminus \Gamma_{C_\xi}$. Then, at every stage ξ reached, limit stages included:*

- (i) C_ξ is convex, and $x + C_\xi \mapsto \Phi_{E_\xi}(P_{E_\xi}(x))$ is an ordered-group isomorphism of A/C_ξ onto the initial group $\mathcal{F}_{E_\xi}(A)$;
- (ii) for finite ξ this is the realization produced by ξ applications of Theorem C, as in corollary 8.4;
- (iii) for $\xi < \eta$ with $E_\eta \neq \emptyset$, the realization at stage η is obtained from that at stage ξ by compressing the interval $b_{E_\xi}[E_\eta]$: $\Phi_{E_\eta} = \Phi_{b_{E_\xi}[E_\eta]} \circ \Phi_{E_\xi}$ on normal forms supported in E_η .

No hypothesis beyond the initiality of A is needed at a limit stage.

Proof. (i) The preimage of a convex subgroup under the order-preserving quotient map by a convex subgroup is convex, and the union of a chain of convex subgroups is a convex subgroup; so every C_ξ is convex, and proposition 11.7 and theorems 11.5 and 13.1 give the realization, with $\mathcal{F}_\emptyset(A) = \{0\}$ when E_ξ is empty. (iii) is corollary 13.3 for $E_\eta \subseteq E_\xi$.

(ii) By induction on finite k . For $k = 0$, $E_0 = \Gamma$ is initial, so every position is retained and Φ_Γ is the identity. Suppose the stage- k realization is $B_k = \mathcal{F}_{E_k}(A)$, an initial group with exponent class $b_{E_k}[E_k]$. The convex subgroup $C_{k+1}/C_k \cong \mathbb{Z}$ corresponds by proposition 11.7 to the interval $\Gamma_{C_{k+1}} \setminus \Gamma_{C_k}$, which is nonempty and cannot contain two points (their monomials would be Archimedean inequivalent in a

cyclic group); so it is the single point $\min E_k$, and $E_{k+1} = E_k \setminus \{\min E_k\}$. The least positive cyclic subgroup of B_k is the image of C_{k+1}/C_k , with bottom exponent $b_{E_k}(\min E_k)$. By proposition 13.5, applied to B_k , Theorem C realizes $B_k/(\text{that subgroup})$ by compressing $b_{E_k}[E_k]$ minus its bottom point, which is $b_{E_k}[E_{k+1}]$. By (iii) the result is $\mathcal{F}_{E_{k+1}}(A)$. $\square$

This answers question 20.2: the limit stage needs no hypothesis, its quotient is realized directly from the cut Γ_{C_λ} rather than as a limit of earlier realizations, and it is coherent with every earlier stage. The recursion itself uses no choice: each least positive element is unique.

14 The residue obstruction for Presburger groups

14.1 The bottom cyclic group and coherent residues

Let G be a $\mathbb{Z}$ -group with least positive element e . The subgroup $\mathbb{Z}e$ is convex: finite induction shows that the interval $[0, me]$ contains exactly $0, e, \dots, me$ for every ordinary $m \in \mathbb{N}$ (as in the proof of corollary 8.6). Its quotient is torsion free, as every ordered abelian group is. In particular $\mathbb{Z}e$ is pure in G :

$$ny \in \mathbb{Z}e \implies y \in \mathbb{Z}e.$$

Lemma 14.1 (Residues). *For each ordinary $n \geq 1$ and $x \in G$ there is a unique $r_n(x) \in \{0, \dots, n-1\}$ such that $x - r_n(x)e \in nG$. The resulting homomorphisms $G \rightarrow \mathbb{Z}/n\mathbb{Z}$ are compatible and define res_G . Moreover*

$$\ker \text{res}_G = \bigcap_{n \geq 1} nG.$$

Proof. Divisibility of $G/\mathbb{Z}e$ gives $x = ny + ke$ for some $y \in G$ and $k \in \mathbb{Z}$. Reduce k modulo n . If two remainders worked, then $(r - r')e = ny$. Purity puts y in $\mathbb{Z}e$, so n divides $r - r'$ and the chosen range forces equality. Addition and compatibility under divisibility of moduli follow from the same uniqueness. The kernel identity is immediate from the zero remainder condition. $\square$

Lemma 14.2 (The divisible kernel, classical). *The subgroup $G^{\text{div}} := \ker \text{res}_G$ is divisible and is the largest divisible subgroup of G .*

Proof. Let $a \in G^{\text{div}}$ and fix $k \geq 1$. Choose $b \in G$ with $kb = a$. For every $n \geq 1$, the membership $a \in knG$ gives $c \in G$ such that $a = knc$. Torsion freeness yields $b = nc$, so $b \in nG$ for all n . Thus $b \in G^{\text{div}}$. Any divisible subgroup is contained in every nG , and therefore in G^{div} . $\square$

This is the divisible-kernel part of [10, Proposition 4.1 and Remark 4.2]; the elementary proof is included to make the subsequent equivalences explicit.

Proposition 14.3 (Split bottom versus ordinary residues). *For a $\mathbb{Z}$ -group G , the following are equivalent:*

- (i) $\mathbb{Z}e$ is a direct summand of G ;
- (ii) $\text{res}_G(G) = \mathbb{Z} \subseteq \widehat{\mathbb{Z}}$;
- (iii) $G = G^{\text{div}} \oplus \mathbb{Z}e$.

In case (iii), the order is $G^{\text{div}} \xrightarrow{\sim} \mathbb{Z}$. Every group complement to $\mathbb{Z}e$ equals G^{div} .

Proof. Assume $G = V \oplus \mathbb{Z}e$. The quotient map identifies V with $G/\mathbb{Z}e$, so V is divisible and $V \subseteq G^{\text{div}}$. For $v + me$ its coherent residue is m . Hence $\text{res}_G(G) = \mathbb{Z}$, and its kernel is exactly V .

Conversely, if every coherent residue is an ordinary integer, then for each x there is $m \in \mathbb{Z}$ such that $\text{res}_G(x) = m$. It follows that $x - me \in G^{\text{div}}$. The intersection $G^{\text{div}} \cap \mathbb{Z}e$ is zero, since no nonzero

ordinary integer is divisible by every positive ordinary integer. Thus (iii) holds, and implies (i). Convexity of $\mathbb{Z}e$ gives the lexicographic ordering for any complement: a nonzero positive quotient lift is larger than every member of $\mathbb{Z}e$. The first argument also proves uniqueness of that complement. $\square$

14.2 Initial omnific realization in the split case

Recall $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ from (4) and section 1.4. The symbol Π denotes purely *infinite* forms, with zero included, not positive elements and not infinitesimals. Source 02 uses the integer-bottom suspension $J(B) = \mathbb{Z} \oplus \sigma_*[B]$ of theorem 8.5: for initial $B \subseteq \mathbf{No}$, $J(B)$ is an initial subgroup of $\mathbf{Oz}$ order-isomorphic to $B \overrightarrow{\times} \mathbb{Z}$. It re-proves this from imported fact 2.2 by the argument of the first paragraph of the proof of theorem 8.5, so that Theorem E does not depend on source 01's separate normalization proof. The statement and proof are printed once, there.

Proof of Theorem E. If $G \in \text{Init}$, convex splitting (theorem 11.5) applied to $\mathbb{Z}e$ in an initial realization makes the bottom cyclic group a direct summand. Proposition 14.3 proves the algebraic and residue equivalences. Conversely, in the split case G^{div} is divisible and ordered. Choose an initial realization B of it by imported fact 11.1. The suspension $J(B)$ is an initial omnific realization of G . An initial subgroup of $\mathbf{Oz}$ is, by the meaning of initial here, an initial subgroup of $\mathbf{No}$. $\square$

Source 01's relative criterion gives the same answer by a second route. For a $\mathbb{Z}$ -group the quotient $G/\mathbb{Z}e$ is divisible, hence initially realizable by imported fact 11.1, so corollary 8.6 reduces initial realizability of G to the splitting of $\mathbb{Z}e$: the first and third conditions of Theorem E.

Corollary 14.4 (Rigidity on the initial side). *The only initially realizable rigid $\mathbb{Z}$ -group is $\mathbb{Z}$ itself.*

Proof. In $G^{\text{div}} \overrightarrow{\times} \mathbb{Z}$, the map $(d, m) \mapsto (2d, m)$ is an ordered-group automorphism, because G^{div} is divisible and torsion free. It is nontrivial unless $G^{\text{div}} = 0$. The standard group $\mathbb{Z}$ has no nonidentity order-preserving group automorphism. $\square$

15 An explicit factorial family of nonsplit models

15.1 Presentation inside an ordinary rational plane

Factorial moduli are cofinal for divisibility, so a profinite integer $\zeta \in \widehat{\mathbb{Z}}$ is specified by its compatible residues modulo $n!$. Let $S_n = S_n(\zeta)$ be the unique integer with

$$0 \leq S_n < n!, \quad S_n \equiv \zeta \pmod{n!}, \quad n \geq 1.$$

Then $S_1 = 0$ and

$$S_{n+1} = S_n + a_n n!, \quad a_n \in \{0, 1, \dots, n\}. \quad (31)$$

In the lexicographically ordered rational plane $\mathbb{Q} \overrightarrow{\times} \mathbb{Q}$, put

$$e = (0, 1), \quad x_n = \left(\frac{1}{n!}, \frac{S_n}{n!} \right), \quad G_\zeta = \bigcup_{n \geq 1} (\mathbb{Z}x_n \oplus \mathbb{Z}e). \quad (32)$$

The groups in the union are nested because

$$(n+1)x_{n+1} = x_n + a_n e. \quad (33)$$

Proposition 15.1 (Structure of the factorial family). *The group G_ζ is countable, has least positive element e , and first-coordinate projection π_1 gives an exact sequence of ordered groups*

$$0 \longrightarrow \mathbb{Z}e \longrightarrow G_\zeta \xrightarrow{\pi_1} \mathbb{Q} \longrightarrow 0. \quad (34)$$

Consequently it is a $\mathbb{Z}$ -group. Its first generator has residue $\text{res}_{G_\zeta}(x_1) = -\zeta$.

Proof. Each stage is a free rank-two group, and the union is countable. Its first-coordinate image is $\bigcup_n (n!)^{-1}\mathbb{Z} = \mathbb{Q}$. If $kx_n + me$ has first coordinate zero, then $k = 0$; hence the kernel is exactly $\mathbb{Z}e$. A positive element of nonzero first coordinate exceeds every integer multiple of e , and the positive elements of the kernel are $e, 2e, \dots$. This proves discreteness and convexity. The quotient is $\mathbb{Q}$, hence divisible. Finally, $n!x_n = x_1 + S_n e$, so $x_1 \equiv -S_n e \pmod{n!G_\zeta}$. Compatibility over factorial moduli gives the residue assertion. $\square$

Theorem 15.2 (Exact splitting test). *For $\zeta \in \widehat{\mathbb{Z}}$,*

$$G_\zeta \in \text{Init} \iff (34) \text{ splits} \iff \zeta \in \mathbb{Z}.$$

Proof. A retraction $r : G_\zeta \rightarrow \mathbb{Z}$ with $r(e) = 1$ is determined by the integers $b_n = r(x_n)$. The relations imply

$$(n+1)b_{n+1} = b_n + a_n, \quad n!b_n = b_1 + S_n. \quad (35)$$

Thus $b_1 + \zeta$ is zero modulo every $n!$, so $\zeta = -b_1 \in \mathbb{Z}$. Conversely, when $\zeta = m \in \mathbb{Z}$, put $b_n = (S_n - m)/n!$. These are integers satisfying all transition relations. On each stage define $r(kx_n + le) = kb_n + l$; compatibility makes this a well-defined retraction on the union. The equivalence with initial realizability follows from Theorem E. $\square$

15.2 A completely explicit noninteger parameter

Define

$$\zeta_* = \sum_{k=1}^{\infty} k! \quad \text{in } \widehat{\mathbb{Z}}, \quad S_n = \sum_{k=1}^{n-1} k!. \quad (36)$$

For each fixed modulus, all sufficiently late factorials vanish, so this is a well-defined profinite sum. It is not an ordinary convergent real series and is not a Hahn sum of omnific integers.

Lemma 15.3. *One has $0 \leq S_n < n!$, $S_n \rightarrow +\infty$, and $S_n/n! \rightarrow 0$ in the ordinary real sense. In particular $\zeta_* \notin \mathbb{Z}$.*

Proof. The first inequality is direct for $n = 1, 2$. If $S_n < n!$, then $S_{n+1} = S_n + n! < 2n! \leq (n+1)!$ for $n \geq 1$, with the small equality case harmless because the first inequality is strict. For $n \geq 3$, induction gives $S_n \leq 2(n-1)!$: it holds at $n = 3$, and

$$S_{n+1} \leq 2(n-1)! + n! \leq 2n! \quad (n \geq 2).$$

Hence $S_n/n! \leq 2/n$. Unboundedness follows from the summand $(n-1)!$. If ζ_* were a nonnegative ordinary integer m , its least residues modulo all sufficiently large $n!$ would be the fixed value m , contrary to unboundedness. If it were $-m$ with $m > 0$, those residues would be $n! - m$, whose ratio to $n!$ tends to one, contrary to the proved ratio limit. $\square$

For this parameter all digits a_n equal 1, and $(n+1)x_{n+1} = x_n + e$. The first generators are shown below. Thus the abstract obstruction has a presentation entirely in terms of pairs of ordinary rational numbers.

n	$n!$	S_n	x_n
1	1	0	(1, 0)
2	2	1	(1/2, 1/2)
3	6	3	(1/6, 1/2)
4	24	9	(1/24, 3/8)
5	120	33	(1/120, 11/40)
6	720	153	(1/720, 17/80)
7	5040	873	(1/5040, 97/560)

Remark 15.4 (Every finite retraction problem is solvable). Fix N . Taking $b_1 = -S_N$ makes $b_n = (S_n - S_N)/n!$ integral for every $n \leq N$ and satisfies every transition visible through stage N . There is nevertheless no single integer b_1 satisfying all stages. The obstruction is precisely the failure of this infinite compatibility problem to have a solution in $\mathbb{Z}$, although it has the profinite solution $-\zeta_*$.

15.3 Extension classes and rigidity

Proposition 15.5 (Classical extension parameter, explicit form). *There is an isomorphism $G_\zeta \rightarrow G_\eta$ inducing the identity on both endpoints of (34) if and only if $\zeta - \eta \in \mathbb{Z}$.*

Proof. Such a map must send e to e and x_n^ζ to $x_n^\eta + b_n e$ with $b_n \in \mathbb{Z}$. Comparing $n!$ times these elements gives

$$n!b_n = b_1 + S_n(\zeta) - S_n(\eta).$$

Thus $b_1 + \zeta - \eta = 0$ in $\widehat{\mathbb{Z}}$. Conversely, if this equality holds for an ordinary integer b_1 , the displayed formula defines integers b_n satisfying the transition relations. They define compatible isomorphisms on the stages and hence an isomorphism on the union; replacing b_n by $-b_n$ gives the inverse. $\square$

This is the concrete presentation behind the familiar parameter $\widehat{\mathbb{Z}}/\mathbb{Z}$. No new computation of the functor Ext is being claimed. For completeness, arbitrary extensions of $\mathbb{Q}$ by $\mathbb{Z}$ admit the same kind of presentation: choose lifts of $1/n!$, adjust their differences by integers, and collect the resulting compatible residues. Only the explicit family and its equivalence relation are needed here.

Theorem 15.6 (Continuum many countable rigid countermodels). *There are $2^{\aleph_0}$ pairwise nonisomorphic countable rigid $\mathbb{Z}$ -groups not in Init , among the G_ζ .*

Proof. The profinite integers have cardinality $2^{\aleph_0}$, and each coset modulo $\mathbb{Z}$ is countable. Proposition 15.5 therefore gives continuum many endpoint-marked extension classes. An ordered-group isomorphism must preserve the least positive element and induces an order-preserving automorphism of $\mathbb{Q}$. Every such automorphism is multiplication by an element of $\mathbb{Q}_{>0}$, a countable set of choices. Thus one ordered-group isomorphism class can account for at most countably many of the marked extension classes. There remain continuum many ordered-group types, with only the zero extension class split.

Let $\zeta \notin \mathbb{Z}$. If G_ζ^{div} were nonzero, its image in $\mathbb{Q}$ would be a nonzero divisible subgroup, hence all of $\mathbb{Q}$. The intersection $G_\zeta^{\text{div}} \cap \mathbb{Z}e$ is zero. Consequently the restriction of π_1 would be an isomorphism $G_\zeta^{\text{div}} \rightarrow \mathbb{Q}$, providing a section and contradicting theorem 15.2. Hence $G_\zeta^{\text{div}} = 0$ and res_{G_ζ} is injective. Every ordered-group automorphism preserves e and all its residues, so injectivity of the residue map forces that automorphism to be the identity. $\square$

The final argument is the residue-injectivity rigidity mechanism studied in [10]. Here it also supplies a particularly explicit family on the noninitial side of Theorem E.

16 Elementary equivalence, dense models, and omitted types

16.1 Initial realizability is not elementary

Corollary 16.1. *Initial realizability is not preserved by elementary equivalence, even among countable $\mathbb{Z}$ -groups. It is not first-order axiomatizable as a property of ordered abelian groups.*

Proof. The standard group $\mathbb{Z}$ is initial. Every G_ζ is a $\mathbb{Z}$ -group, hence elementarily equivalent to $\mathbb{Z}$ by the characterization of Presburger models [10]. For $\zeta \notin \mathbb{Z}$ it is not initially realizable. A first-order axiomatizable class is closed under elementary equivalence. $\square$

The same contrast already occurs with the explicit initial nonstandard $\mathbb{Z}$ -group $\mathbb{Q}\omega + \mathbb{Z}$, order-isomorphic to $\mathbb{Q} \overrightarrow{\times} \mathbb{Z}$. It has the same Presburger theory as every nonsplit G_ζ .

Theorem 16.2 (Dense elementary-equivalence counterexample). *For $\zeta \notin \mathbb{Z}$, let $\tilde{G}_\zeta = G_\zeta \overrightarrow{\times} \mathbb{Q}$. Then $\tilde{G}_\zeta$ is countable and densely ordered, is not initially realizable, and is elementarily equivalent as an ordered group to the initial subgroup*

$$W = \mathbb{Q}\omega^2 + \mathbb{Z}\omega + \mathbb{Q} \subseteq \mathbf{No}.$$

Proof. The final $\mathbb{Q}$ layer is convex and densely ordered. Between two elements with distinct upper coordinates one can increase the lower coordinate of the smaller element; with equal upper coordinates one uses density of $\mathbb{Q}$. Thus the whole group is densely ordered. The quotient by the bottom copy of $\mathbb{Q}$ is G_ζ . If $\tilde{G}_\zeta$ were initially realizable, convex-factor closure would make G_ζ initially realizable, a contradiction.

Since $G_\zeta \equiv \mathbb{Q} \overrightarrow{\times} \mathbb{Z}$, adjoining a disjoint second sort interpreted as the fixed ordered group $\mathbb{Q}$ preserves elementary equivalence. This can be verified by induction on formulas: formulas in disjoint sorts reduce to Boolean combinations of formulas in the individual sorts. The Cartesian product with coordinatewise addition and lexicographic order is uniformly interpretable in that two-sorted structure. It follows that

$$\tilde{G}_\zeta \equiv (\mathbb{Q} \overrightarrow{\times} \mathbb{Z}) \overrightarrow{\times} \mathbb{Q} \cong W.$$

Finally, the exponent class $\{0, 1, 2\}$ is initial and has no right ancestors internally, and the coefficient groups $\mathbb{Q}, \mathbb{Z}, \mathbb{Q}$ are initial real groups. Finite support gives truncation closure and cross-sectionality, so the criterion proves that W is initial. $\square$

16.2 A single recursive type witnessing the obstruction

With the computable integers S_n of (36), consider the partial type in one variable

$$\tau_*(X) = \{X \equiv S_n e \pmod{n!G} : n = 1, 2, \dots\}. \quad (37)$$

Each congruence abbreviates the first-order formula $\exists Y (X = n!Y + S_n e)$ in the ordered-group language with e named. The least positive element is itself definable, so naming it does not change the substantive distinction.

Theorem 16.3 (An omitted computable type). *The type τ_* is finitely satisfiable in every $\mathbb{Z}$ -group and omitted by every initially realizable $\mathbb{Z}$ -group. It is a recursive type under a fixed standard coding of these canonical formulas. Consequently no recursively saturated $\mathbb{Z}$ -group is initially realizable.*

Proof. A finite set of congruences is satisfied by $S_N e$ for any sufficiently large N , because $S_N \equiv S_n \pmod{n!}$ for $n \leq N$. A realization of all congruences has coherent residue ζ_* . That residue is not in $\mathbb{Z}$, by lemma 15.3, whereas Theorem E requires the entire residue image to be $\mathbb{Z}$. Thus the type is omitted in every initially realizable $\mathbb{Z}$ -group.

The sequences $n!$ and S_n are computable. Using canonical integer numerals, the set of displayed formulas is decidable: read the modulus, decide whether it is a factorial, and check the corresponding residue, handling the duplicate modulus 1 at the start by our convention $n \geq 1$. Standard definitions of recursive saturation therefore apply. A recursively saturated model realizes every finitely satisfiable recursive type over finitely many parameters, contradicting the omission. $\square$

Corollary 16.4 (Countable saturation). *If a $\mathbb{Z}$ -group is countably saturated, its residue map is surjective onto $\widehat{\mathbb{Z}}$. In particular it is not initially realizable.*

Proof. For an arbitrary $\zeta \in \widehat{\mathbb{Z}}$, use its compatible residues $S_n(\zeta)$ instead of S_n in (37). The resulting countable type is finitely satisfiable by ordinary multiples of e . Countable saturation supplies a realization. Taking ζ_* yields the final assertion without any assumption on the continuum hypothesis. $\square$

Theorem 16.5 (Every nonprincipal sequential ultrapower fails). *Let G be any $\mathbb{Z}$ -group and U any nonprincipal ultrafilter on $\mathbb{N}$. Then $G^{\mathbb{N}}/U$ is not initially realizable.*

Proof. By Łoś's theorem it is a $\mathbb{Z}$ -group. In the ultrapower take $a = [(S_N e)_{N \geq 1}]_U$. For any fixed n , the congruence $S_N e \equiv S_n e \pmod{n!G}$ holds for all $N \geq n$, hence on a set in U . Therefore a realizes τ_* . The previous theorem rules out an initial realization. The argument uses only cofinite sets and does not invoke an ultrapower saturation theorem. $\square$

16.3 An obstruction for nonstandard Peano models

Theorem 16.6 (Nonstandard arithmetic cannot have an initial additive realization). *Let M be a nonstandard model of Peano arithmetic, and let $G(M)$ be the ordered abelian group completion of its additive monoid. Then $G(M)$ is not initially realizable in **No**.*

Proof. The ordinary division algorithm, as proved in Peano arithmetic, gives unique division with remainder by every positive standard integer. Consequently $G(M)$ is a $\mathbb{Z}$ -group with least positive element 1.

Choose a nonstandard $h \in M$. Peano arithmetic defines the factorial function and the primitive-recursive partial sums

$$F(h) = \sum_{k=1}^h k!.$$

The sum here is the internally coded finite sum in M , not an external infinite group sum. For each fixed ordinary n , arithmetic proves that $n!$ divides $k!$ whenever $k \geq n$, and therefore proves

$$h \geq n \implies F(h) \equiv S_n \pmod{n!}.$$

This follows by induction on the finite-sum length inside the model; all the displayed factorial and summation recursions are provably total in Peano arithmetic. Since h exceeds every standard n , $F(h)$ realizes the external type τ_* in $G(M)$. Theorem 16.3 excludes initial realizability. $\square$

The arithmetic infrastructure used here is standard; factorials and division can in fact be handled in much weaker arithmetic, as noted in [12]. The fragment is not optimized here. The conclusion is stronger than merely observing that the usual omnific multiplication does not make the positive cone of **Oz** a nonstandard Peano model: it excludes *any* such model having an initially realizable ordered additive group.

Proof of Theorem F. The discrete family and rigidity are theorem 15.6 and Presburger elementary equivalence. The dense examples are theorem 16.2. The remaining assertions are theorems 16.3, 16.5 and 16.6. $\square$

17 Omnific and surcomplex consequences

17.1 The precise residue image of omnific integers

Proposition 17.1. *The additive group of **Oz** has least positive element 1, its purely infinite subgroup Π is divisible, and for every ordinary $n \geq 1$*

$$\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}, \quad \text{res}_{\mathbf{Oz}}(z) = \text{ct}(z) \in \mathbb{Z} \subseteq \widehat{\mathbb{Z}}.$$

The type τ_* is finitely satisfiable in $\mathbf{Oz}$ and has no realization there.

Proof. The normal form (4) gives $z = y + m$ uniquely with $y \in \Pi$ and $m \in \mathbb{Z}$. Every nonzero purely infinite form dominates all real constants in absolute value, proving the least-positive claim. For a positive ordinary n , dividing the real coefficients of y by n leaves its support strictly positive. Thus Π is divisible and $z \equiv m \pmod{n\mathbf{Oz}}$. The kernel of the residue map to $\mathbb{Z}/n\mathbb{Z}$ is exactly $n\mathbf{Oz}$, proving both formulas. The final assertion follows by the same explicit residue argument as theorem 16.3. $\square$

This is a class-function statement; it does not require a set of all omnific integers or a set quotient by a proper-class subgroup. The finite quotient is represented directly by its ordinary remainder.

Corollary 17.2 (Unit-preserving omnific embeddings). *For a set-sized $\mathbb{Z}$ -group G with least positive element e , the following are equivalent: $G \in \text{Init}$; there is an order-preserving additive embedding $f : G \rightarrow \mathbf{Oz}$ with $f(e) = 1$; there is an additive homomorphism $f : G \rightarrow \mathbf{Oz}$ with $f(e) = 1$.*

Proof. An initial omnific realization supplies the stated embedding and takes the least positive element to 1. Conversely, for any such homomorphism, $\text{ct} \circ f : G \rightarrow \mathbb{Z}$ is a retraction on $\mathbb{Z}e$. Thus the bottom cyclic group splits, and Theorem E applies. $\square$

In particular, the ordered additive completion of a nonstandard Peano model cannot even admit a unit-preserving additive embedding into $\mathbf{Oz}$, whether or not the image is initial. There is no contradiction with ordinary ordered-group embeddings that move the distinguished unit: the copy $(a, b) \mapsto a\omega^2 + b\omega$ embeds each G_ζ into $\mathbf{Oz}$ but sends e to ω , not to 1.

17.2 Period groups of surcomplex phases

The trigonometry report studies the normalized period group $I(\Psi)$ of a phase Ψ on a field F (`trigonometry:per:def:periodgroup`). It is an additive integer part of F with least positive element 1, it sits in the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow I \rightarrow \Pi_F \rightarrow 0$ given by the principal part (`trigonometry:per:thm:dictionary`), and for set-sized F it is a $\mathbb{Z}$ -group (after `trigonometry:per:prop:division`). Its residue map res_I is the map res_G of lemma 14.1 for $G = I$, $e = 1$ (`trigonometry:per:eq:residue`), and its profinite defect $\partial_\Psi : \Pi_F \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ (`trigonometry:per:thm:defect`) is that residue taken on a lift and read modulo $\mathbb{Z}$. Source 02's criterion therefore decides which period groups are initially realizable. The statement is the merge's.

Corollary 17.3 (Initially realizable period groups; [merge]). *Let F be $\mathbf{No}$ or one of the set-sized fields of `trigonometry:per:sec:phases`, and let Ψ be a phase on F with normalized period group $I = I(\Psi)$, ordered as a subgroup of F . The following are equivalent:*

- (i) I is initially realizable;
- (ii) the profinite defect ∂_Ψ vanishes;
- (iii) the period sequence $0 \rightarrow \mathbb{Z} \rightarrow I \rightarrow \Pi_F \rightarrow 0$ splits.

In that case $I \cong \Pi_F \overrightarrow{\times} \mathbb{Z}$ as ordered groups; for $F = \mathbf{No}$ and for the initial fields $\mathbf{No}(\lambda)$ this is an order-isomorphism onto the canonical period group $\mathbf{Oz} \cap F$, which is initial.

Proof. (ii) $\Leftrightarrow$ (iii) is `trigonometry:per:thm:split`, conditions (1) and (2).

(i) $\Rightarrow$ (iii). The element 1 is the least positive element of I and $I \cap \mathcal{O}_F = \mathbb{Z}$. An element of I outside $\mathbb{Z}$ is infinite, so its sign is the sign of its principal part; hence the principal part induces an ordered isomorphism $I/\mathbb{Z} \cong \Pi_F$. If I is initially realizable, the convex subgroup $\mathbb{Z}$ is a direct summand by theorem 11.5 (or by corollary 8.6), and the period sequence splits.

(iii) $\Rightarrow$ (i). A splitting gives $\mathcal{I} = V \oplus \mathbb{Z}$ with V mapped isomorphically onto Π_F by the principal part. By convexity of $\mathbb{Z}$ the order is lexicographic, and since the principal part preserves signs on V , the map $v + m \mapsto \text{pr}(v) + m$ is an order-isomorphism $\mathcal{I} \rightarrow \Pi_F \oplus \mathbb{Z}$, the latter ordered as a subgroup of F , which is $\Pi_F \overrightarrow{\times} \mathbb{Z}$. For $F = \mathbf{No}$ the target is $\Pi \oplus \mathbb{Z} = \mathbf{Oz}$, and for $F = \mathbf{No}(\lambda)$ it is $\mathbf{Oz} \cap F$; both are initial ($\mathbf{Oz}$ by (4), and an intersection of initial classes is initial). For a set-sized F , Π_F is a divisible ordered group, so it has an initial realization B by imported fact 11.1, and $J(B)$ realizes $\Pi_F \overrightarrow{\times} \mathbb{Z}$ (theorem 8.5).

Second route for set-sized F : $\mathcal{I}$ is a set-sized $\mathbb{Z}$ -group with $\text{res}_{\mathcal{I}}(\mathcal{I}) = q^{-1}(\partial_{\Psi}(\Pi_F))$, q the quotient map $\widehat{\mathbb{Z}} \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ (`trigonometry:per:thm:defect`); this is $\mathbb{Z}$ exactly when $\partial_{\Psi} = 0$, and Theorem E gives (i) $\Leftrightarrow$ (ii). $\square$

So a period group is initially realizable exactly when its profinite defect is zero. The corollary is additive: it says nothing about multiplicative closure of the periods, which the trigonometry report shows to be independent of the defect (`trigonometry:per:sec:tests`), and it does not make a noncanonical split period group equal to $\mathbf{Oz} \cap F$ as a subclass. The nonsplit period groups are further instances of the obstruction of Theorem E.

17.3 Rectangular surcomplex factors

Let $\mathbf{No}[i]$, $A[i]$ and coordinatewise initiality be as in section 9 (definition 9.1). This is an explicitly coordinate-dependent notion; no single canonical sign-tree order on $\mathbf{No}[i]$ is postulated.

Theorem 17.4 (Rectangular convex-factor transport). *Let $G \subseteq \mathbf{No}$ be initial and $C \subseteq C'$ convex in G . For the interval $E = \Gamma_{C'} \setminus \Gamma_C$, the map*

$$a + ib \mapsto \Phi_E(P_E a) + i\Phi_E(P_E b), \quad a, b \in C',$$

induces a $\mathbb{Z}[i]$ -linear isomorphism

$$C'[i]/C[i] \cong \mathcal{F}_E(G)[i].$$

It commutes with conjugation. The target is coordinatewise initial, and the ambient complex-coefficient Hahn transport preserves and reflects set-indexed strong summability.

Proof. The real factor map has kernel C in each coordinate. The induced map is therefore bijective on the rectangular quotient. Additivity and compatibility with $(a, b) \mapsto (-b, a)$ give $\mathbb{Z}[i]$ -linearity, and negating the imaginary coordinate gives conjugation compatibility. Initiality of the real target is theorem 13.1. A complex normal form is obtained by the union of the two real supports, still reverse well ordered. Reindexing changes neither that order nor the finite set of contributors at any exponent, so the strong-sum proof is the same coefficientwise proof as in the real case. $\square$

Proposition 17.5 (Gaussian omnific residues). *For each ordinary $n \geq 1$,*

$$\mathbf{Oz}[i]/n\mathbf{Oz}[i] \cong \mathbb{Z}[i]/n\mathbb{Z}[i], \quad \text{res}_{\mathbf{Oz}[i]}(a + ib) = \text{ct}(a) + i \text{ct}(b) \in \mathbb{Z}[i] \subseteq \widehat{\mathbb{Z}}[i].$$

Proof. Apply proposition 17.1 to both coordinates. Multiplication by i interchanges them with a sign, so the coordinate residue map is $\mathbb{Z}[i]$ -linear. Its kernel modulo n is $n\mathbf{Oz}[i]$. $\square$

These are additive and module-theoretic conclusions. The exponent compression need not respect addition of exponents, so it does not in general preserve products of normal forms. No field, ring, differential, or exponential equivalence is inferred from theorem 17.4.

18 Boundaries of the convex-factor theory

18.1 Why these conclusions do not classify every ordered group

Convex splitting is necessary for initial realizability, but no sufficiency theorem for that condition alone is proved here. Even a truncation-closed cross-sectional Hahn presentation must meet the coefficient and right-ancestor conditions. The proof of Theorem E succeeds because the remaining quotient is divisible and hence has an initial realization by an established theorem. For a general discrete group that quotient need not be divisible.

Arbitrary subgroup closure is false. In fact the rational-coordinate map $(a, b) \mapsto a\omega + b$ embeds every G_ζ into the initial group $\mathbb{Q}\omega + \mathbb{Q}$. For $\zeta \notin \mathbb{Z}$ this gives a concrete noninitially-realizable subgroup of a two-level initial group. The structural theorem requires convex factors or separately verified coordinate restrictions. Likewise, a convex subgroup may fail literal initiality in its original position. In the initial group $\mathbb{D} + \mathbb{Z}\omega^{-1}$ of section 10, the subgroup $\mathbb{Z}\omega^{-1}$ is convex and does not contain the simpler number 1. It nevertheless has the initial realization $\mathbb{Z}$, as the theorem predicts.

18.2 The full optimality question is still separate

Ehrlich and Kaplan prove an optimality theorem allowing class models, for theories between ordered abelian groups and divisible ordered abelian groups, and ask for a set-model counterpart (Section 8 of [1]: Theorem 6 and Question 1; in the journal version, Theorem 8.1 and Question 8.1 [2]). The explicit dense groups of theorem 16.2 are set-sized noninitial examples, but are not a construction uniform over every proper intermediate theory.

In particular, Presburger arithmetic asserts discreteness and is not a subtheory of divisible ordered abelian groups. Thus Theorem E and theorem 15.6 alone cannot answer their full question by simply substituting Presburger arithmetic for an intermediate theory. The dense variant in theorem 16.2 removes discreteness of the whole group, but still does not establish the required uniformity over all intermediate theories. Source 02 makes no claim to have resolved that published question in its entirety.

Source 02 answers none of Ehrlich and Kaplan’s questions. Question 2 (journal 9.1) is the subject of source 01 (corollary 6.2), and source 02 does not use that answer: from source 01 it takes only the suspension, which it re-proves. Question 3 (journal 9.2), the initial-monoid problem, is not touched. Question 1 (journal 8.1) is the one discussed here.

Remark 18.1 (The two formulations of the optimality theorem; [merge]). The arXiv version quantifies over theories T with $T_{\text{OAG}} \subseteq T \subseteq T_{\text{DOAG}}$; since $\text{Th}(W)$ proves non-divisibility, it is not such a theory. The journal version, read at placement in the authors’ text hosted on the second author’s web page, states Theorem 8.1(i) for theories in the language $\{<, +, 0\}$ containing the theory T_D of nontrivial densely ordered abelian groups, and Question 8.1 asks whether its “only if” part can be established in NBG with set models alone. In that formulation $\text{Th}(W)$ is in scope, and theorem 16.2 gives, for every theory $T \supseteq T_D$ in that language that holds in W , a countable model of T that is not initially realizable. This settles the set-model “only if” part for those theories only, as Ehrlich and Kaplan’s own example of a p -divisible group does for others; it gives no construction for every theory lacking full divisibility, and the question stays open in both versions.

18.3 Product and completion pitfalls

The group G_ζ for $\zeta \notin \mathbb{Z}$ has both convex kernel $\mathbb{Z}$ and quotient $\mathbb{Q}$ initially realizable, but is not itself initially realizable. Thus closure under convex factors has no unrestricted converse for extensions. The missing datum is the extension class.

By contrast, theorem 11.5 gives an actual complement whenever the middle group is initial. This

complement depends on the normal-form realization; a group automorphism can change a splitting in general. Only the Presburger split case has the uniqueness proved in [proposition 14.3](#).

The maps P_E for infinitely many disjoint intervals may form an ambient strongly summable family when applied to one fixed normal form, but this does not identify an arbitrary initial group with the full Hahn product of its factors. Such an identification would reintroduce the unsupported completion assumption excluded in [remark 11.2](#).

19 Verification, provenance, and a formalization route

Each subsection below first gives source 01's record, as delivered, and then source 02's; "this manuscript" in source 01's text means source 01.

19.1 The exact dependency chain

The proof separates into three kinds of input. Standard surreal normal forms identify the ambient additive spaces. The established Ehrlich–Kaplan criterion decides when a specified canonical Hahn image is initial. Everything connecting the original discrete group to the normalized image is then given by the explicit formulas and proofs in this manuscript.

The essential chain is

$$\begin{aligned} & \text{bottom coefficient structure} \longrightarrow \text{root relocation} \\ & \longrightarrow \text{strong Hahn transport} \longrightarrow \text{initiality of the image.} \end{aligned}$$

For the inverse classification, the additional chain is

$$\begin{aligned} & \text{missing ancestors of } p \longrightarrow \text{spine exponents} \\ & \longrightarrow \text{forced dyadic coefficients} \longrightarrow K_\alpha \subseteq H. \end{aligned}$$

The sign-sequence proof is independent of the arithmetic of exponents. In particular, no unproved assertion that Θ_α preserves addition of exponents occurs in the chain.

Source 02. The convex-factor theorem depends on normal forms and the initial Hahn criterion, then only on the proved projection and sign lemmas. The Presburger criterion adds the divisible initial-embedding theorem ([imported fact 11.1](#)). The explicit nonsplit examples add no further structural input. The elementary-equivalence consequences use the classical characterization of Presburger models; the ultrapower statement additionally uses Łoś's theorem. The Peano statement uses the usual provably total factorial and summation recursions. The surcomplex module assertions are coordinatewise consequences of the real factor theorem. The merge statements [proposition 13.5](#) and [corollaries 13.6](#) and [17.3](#) add source 01's [sections 4](#) and [8](#) and, for the last, statements of the trigonometry report cited by label.

19.2 Finite regression checks supplied with the article

The accompanying Python program `verify.py` (shipped in this report as `code/verify.py`) uses only the standard library, exact rational arithmetic, and a fixed random seed. Its recorded run performs **450,862 assertions**, all passing (shipped as `data/verification_report.json`). The tests cover finite sign words through length seven for all relevant finite cone minima; every prefix-closed subset of the binary tree through depth three, including the empty tree; inverse-spine tests on a further family of nonnegative initial trees; and 1,500 random pairs of finite rational normal forms.

The count of prefix-closed trees is independently specified by

$$T_{-1} = 1, \quad T_d = 1 + T_{d-1}^2,$$

so $T_0 = 2$, $T_1 = 5$, $T_2 = 26$, and $T_3 = 677$. The recurrence counts the empty tree, or a root with independently chosen initial left and right subtrees. This makes the finite-tree coverage exhaustive at the stated depth, rather than a random sample.

Tests check inverse identities, order preservation, the exact predecessor identity, the right-ancestor identity away from the minimum, loss of the minimum's right ancestors, initiality of transported trees, the exact inverse exponent criterion, and the source dyadic obligations recovered by adjoining the spine. The series tests check inverse transport, additivity, order, truncations, and integrality of the bottom coefficient. Separate negative tests reject the false global right-ancestor identity and unrestricted inversion.

The program does not represent arbitrary ordinals, arbitrary real coefficients, or infinite Hahn supports. It does not check the imported characterization theorem or establish absence of a transfinite proof gap. Its role is regression testing of the finite combinatorics and coefficient formulas. The written proofs, not these experiments, carry the general claims.

When this report was placed, the program was rerun on a copy of the directory with an explicit output file. It passed with the same 450,862 assertions, and its JSON output was identical in content to the shipped record. The program's default output file is `verification_report.json` in the current directory, and the delivered build script calls it with that name, so a rerun should always be made on a copy (the collection README gives the commands).

Source 02. Its program, shipped as `code/02-convex-factors-verify.py` (delivered as `code/verify.py`), uses only the Python standard library and exact integers and rational fractions. Its recorded run, shipped as `data/02-convex-factors-verification.json` (delivered as `data/verification.json`), returns PASS. The main coverage is:

Check family	Count
All nonempty convex intervals in the sign tree of lengths ≤ 5	2,016
All nonempty meet-closed subsets of the sign tree of lengths ≤ 2	61
Pairs checked for order, prefix reflection, and meet preservation	1,398,609
Nested convex interval pairs in the sign tree of lengths ≤ 4	46,376
Points checked for two-stage compression coherence	324,632
Factorial transition identities	100
Compatibility checks between factorial residues	5,050
Integer-parameter splitting identities	1,769
Finite projection composition identities	24,300

No row certifies a transfinite theorem. The program additionally checks initiality of every finite compressed image (145,740 prefix checks), the non-meet-closed collision example, the solvability of all finite retraction systems through sixty stages, the factorial bound used in lemma 15.3, and additivity of interval projections (2,970 checks); the full counts are in the JSON record. The successful finite retractions are a deliberate regression case: the code must not mistake finite solvability for a global splitting. The tests do not verify the Ehrlich–Kaplan theorem, arbitrary ordinals, class comprehension, completeness of Presburger arithmetic, recursive saturation, or the Peano-model argument. Those depend on the written proofs and the explicitly imported results, not on the test counts.

The program prints its JSON report and writes a file only when `-output` is given; the delivered README's rerun command names the delivered record as that output and would overwrite it, so it too should be rerun on a copy. At this write it was rerun on a copy with a scratch output file: it passed in about ten seconds, and the JSON content was identical to the shipped record.

19.3 Relationship to the Surreal repository

The repository review examined the public README, the introductory coverage and dependency ledger, and targeted searches for overlap [6]. Those materials describe extensive sign-sequence, canonical cut,

normal-form, Hahn-summation, and algebraic infrastructure. They also explicitly distinguish source manuscripts from checked Lean declarations. The present argument is designed to fit that distinction.

The review took place while the repository was changing. The initially inspected README and ledger had blob hashes

```
391afd7c64cebd48b47fbe43dae26c27e8ee8047
6a89497cb1de3e5c7a346201f7fc006d85082426.
```

A later metadata read reported commit

```
173eb522fdf5d4bf201c097f76b38273eb4f95e1.
```

The initially returned recursive-tree hash is not used as a commit identifier. This manuscript does not claim to have reviewed every repository article or every companion archive, and no repository result is silently promoted from “pending” to “proved.” The manuscript has no pinned commit. The commit 173eb52 above is an ancestor of the commit c6359e4 that placed the manuscript in the collection. The manuscript makes no claim about the content of any report, so no statement of it had to be corrected against the collection as it now stands.

A sound implementation would first formalize the following mathematical interfaces; the names are proposed work units, not declarations claimed to exist:

1. *Cone root relocation.* Define the three-way sign split, the inverse, the order equivalence, and (8) for ordinal-length sequences.
2. *Hahn reindexing with one coefficient multiplier.* Prove support transport, real-linearity, order preservation, truncation compatibility, and strong-sum interchange.
3. *Concrete initial-subgroup criterion.* Supply a checked version of [imported fact 2.2](#), including both its coefficient and right-ancestor conditions, before claiming the final theorem.
4. *Normalization and exact inverse.* Instantiate the criterion for the transformed groups; then derive the spine classification, quotient realization, and complex-coordinate extension.

The largest imported proof obligation is the third item. A checked conditional theorem using that criterion would be valuable, but would not by itself constitute a checked theorem about every actual initial subgroup of **No**. This package supplies no Lean file with that obligation hidden as an axiom or an unfinished proof.

Source 02. Its repository review was made at commit

```
9693b28c24e6fcb317ce47969a40185c5dfef402,
```

and included the main README, the formalization ledger, the article catalogue exposed there, and the guide of the omnific groups and lattices report; these distinguish mathematical drafts from Lean results. It did not constitute a claim-by-claim audit of the repository or a local Lean build. Source 02 read source 01 from the project library, including its quotient, suspension and relative splitting arguments, and recorded that its location in the inspected Git tree was not established, citing it as a preceding project draft without assigning it a repository path. That was correct at its pin: 9693b28 is an ancestor of the commit c6359e4 that placed source 01, and the preceding draft is now [sections 1 to 10](#) of this report (its “Section 8” is [section 8](#)). No theorem of source 02 is called Lean-verified. A sensible implementation order would formalize the finite and transfinite sign-tree compression first, followed by the support projection lemmas and the transfer of the initial Hahn criterion. The residue and factorial-presentation arguments can be formalized in ordinary abelian-group and ordered-group libraries independently of a full implementation of omnific integers. This is a dependency plan, not delivered formal code.

19.4 Novelty and literature status

The new contributions claimed provisionally here are the explicit root-relocation map together with its exact predecessor identity, its use in omnific normalization, and the sharp dyadic-spine image classification. The extremal, quotient, and surcomplex statements are derived refinements of that construction. The known two-level example, the general initial-Hahn-subgroup characterization, and the classification of initial real coefficient groups are not claimed as new.

The literature check recovered the stated question in the original preprint, thesis, and workshop account [1, 3, 4], and examined adjacent sign-concatenation work [5]. Targeted searches did not locate a prior resolution or the specific root-relocation argument. That negative search result is limited evidence: it does not exclude an unindexed paper, unpublished proof, or a reformulation already known to a specialist. Neither mathematical correctness nor priority is certified by the absence of a search result.

Source 02. The convex-factor theorem and its surreal applications are proposed contributions with complete proofs; neither priority nor independent referee certification is claimed. The Ehrlich–Kaplan criterion, Ehrlich’s divisible embedding, Jeřábek’s residue theory and residue-injectivity rigidity mechanism [10], and $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ [11] are classical and credited; no new Ext computation is claimed, and the existence of rigid Presburger models is not claimed as new. Bagayoko and van der Hoeven’s surreal substructures [5] are a related viewpoint, not a dependency, and [12] is background on internal arithmetic only. A targeted source review did not find a prior formulation of the main convex-factor theorem; the search was not exhaustive.

19.5 Provenance, placement review, and place in the collection

Two sources. This report is built from two manuscripts.

Source 01 is *Discrete Initial Groups and Omnific Normalization* (24 pages, 23 September 2026), manuscript 09 of the collection’s batch 28. It was delivered with a README, a source audit, the finite checks `verify.py`, their recorded result and a build script. It was placed in commit `c6359e4` as this directory and written in `3d9dbe9`. Its LaTeX source is the basis of `article.tex`; its source audit is shipped unchanged as `source_audit.md`, the program and build script as `code/verify.py` and `code/build.sh`, and the record as `data/verification_report.json`. Its delivered PDF and README are not shipped. It has no pinned repository commit; its read identifiers are those recorded above. It contributes sections 1 to 10, Appendix A, Questions 20.1–20.4, non-claims (N1)–(N15) and most of the checklist of Appendix B.

Source 02 is *Convex Factors and Profinite Obstructions for Initial Surreal Groups* (23 pages, 23 September 2026), manuscript 09 of the collection’s batch 29. It was delivered with a README, a provenance record, the program `code/verify.py` and its record `data/verification.json`, and placed in commit `66d7e55` under the prefix `02-convex-factors-`: the provenance record is shipped unchanged as `02-convex-factors-PROVENANCE.md`, the program as `code/02-convex-factors-verify.py` and the record as `data/02-convex-factors-verification.json`. Its LaTeX source, PDF and README are not shipped; the provenance record names the delivered file names (`code/verify.py`, `data/verification.json`), which in this directory carry the prefix. Its repository pin is `9693b28`. It contributes section 1.5, sections 11 to 18 except the statements marked [merge], Appendix C, Questions 20.5–20.8, non-claims (N16)–(N29) and the corresponding rows of the checklist.

How the merge was made (source 02). Source 02 was placed as an addition because it answers two questions this report named, questions 20.1 and 20.2. Its material is printed as it stands, with these choices. Results that both sources contain are printed once: the Ehrlich–Kaplan criterion (imported fact 2.2), the classification of initial real groups (imported fact 2.3), the caution against Hahn

completion (remarks 5.3 and 11.2), the integer-bottom suspension (theorem 8.5; source 02 re-proves it from the criterion, by the same argument), the description of **Oz** ((4)) and the rectangular convention (definition 9.1). Source 02's foundations section is replaced by section 2 and section 11.1; its symbols that collide with source 01's are renamed (section 11.1); its Theorems A–C are Theorems D–F. Its verification, repository, novelty, question and non-claim material is merged into the corresponding parts of sections 19 and 20; its notation table is replaced by the renaming table of section 11.1 and the definitions at their places, its dependency summary is in section 19, and its independent-review checks are in Appendix B. Three statements are added in the merge, each with a complete proof and marked [merge]: proposition 13.5 (source 02's construction for $\mathbb{Z}\varepsilon$ is source 01's $\rho \circ \Theta_\alpha$), corollary 13.6 (limit stages of the iteration of corollary 8.4) and corollary 17.3 (period groups of the trigonometry report); remark 18.1 records how the two versions of Ehrlich and Kaplan's optimality theorem differ. The Ehrlich–Kaplan numbering table of section 1 gained the two rows for Section 8. No mathematical statement, proof or example of source 01 was changed. Where source 01 said that convex quotients and limit stages were not covered (after corollaries 8.4 and 8.6, in questions 20.1 and 20.2 and in (N10)), the text now says which source proves what; the title page, abstract, status box, conventions, this section and the conclusion mention the second source.

What the write changed (source 01). Every label now carries the prefix `isg:`. The journal numbering of the Ehrlich–Kaplan results (section 1) and the wording of the journal version of their criterion (after imported fact 2.2) were added, as were section 1.4, this subsection, the shipped file names, a definition of `ct` (which the manuscript used without defining it), labels on formerly unlabelled statements, and the collected list of non-claims (section 20.1). Cross-references now name each kind of statement correctly; in the delivered source every reference to a lemma, proposition, corollary, remark or imported fact printed as “theorem”. No mathematical statement, proof or example was changed.

Placement review (source 01). When source 01 was placed, every step of the argument was checked by hand: the bottom-coefficient structure, both cases of the predecessor identity (8) and its failure at the old minimum, the ambient transport, every dyadic obligation in the proof of Theorem A, both directions of Theorem B with its two counterexamples, the extremal groups, the size and depth corollaries, the splitting and suspension theorems, and every example of section 10. No gap was found. The Ehrlich–Kaplan criterion was checked as a statement, against the arXiv text and the author-hosted text of the journal version: imported fact 2.2 is a faithful reading of [1, Theorem 1 and its proof], equivalently of Theorem 5.1 of [2], and imported fact 2.3 is [1, Lemma 3]. The criterion was not re-proved; the whole argument rests on it. This review is not a refereeing, and the solution proposed here has not been refereed. Its scope is source 01's material; it does not cover sections 11 to 18.

Placement review (source 02). When source 02 was placed, its proofs were checked by hand and no error was found: the exponent-cut lemmas and the canonical splitting, the compression theorem with its uniqueness and nested coherence, the transfer of every dyadic obligation in theorem 13.1 (coefficient groups are unchanged and every target right ancestor comes from a source one by (27)), the residue lemmas and proposition 14.3, the factorial family (its table, the transitions, $n!x_n = x_1 + S_n e$, $S_n \leq 2(n-1)!$ and the finite retractions), the extension classes and rigidity, the dense example (initiality of W : its exponents 0, 1, 2 have no right ancestors in the class, and $\mathbb{Q}$ and $\mathbb{Z}$ are initial in $\mathbb{R}$), the omitted type, the ultrapower and Peano arguments, the omnific residues and the rectangular transport. The compression was re-implemented independently and tested on all 496 nonempty convex intervals of the sign tree of lengths at most four (order, injectivity, initial image, prefix and right-ancestor reflection) and for nested coherence on the tree of lengths at most three. The elementary equivalence in theorem 16.2 rests on a standard two-sorted interpretation, checked as an argument. The Ehrlich–Kaplan criterion and Ehrlich's embedding theorem were not re-proved. For the merge statements, (30) was confirmed by a finite computation for $\alpha \leq 3$ and words of length at most six, and the iteration in corollary 13.6 for

three removal steps on those trees; that computation is not shipped. This review is not a refereeing, and it does not extend to source 01’s material beyond the consistency statements of sections 13.2 and 13.3.

Literature. The placement review of source 01 ran targeted web searches (“discrete initial subgroup”, “initial subgroup” together with “omnific integers”, and the title of [1]). They found only the Ehrlich–Kaplan paper, Kaplan’s thesis and unrelated pages, and no resolution of the question. The search was targeted, not exhaustive. The citations of Kaplan’s thesis (its Sections 6.2 and 7.1) and of the Oberwolfach report (printed pages 3355–3356) are the manuscript’s; they were not re-checked at placement. For source 02 no separate literature search is recorded at placement; its own review was targeted and found no matching formulation of the convex-factor theorem. Its citations of Jeřábek, Strickland, Enayat–Hamkins–Wcisło and Bagayoko–van der Hoeven are the manuscript’s and were not re-checked. Its account of Ehrlich and Kaplan’s Section 8 was compared with the arXiv text and the authors’ text of the journal version at this write (remark 18.1).

Place in the collection. At the placement of source 01, no other report named Question 2 (Question 9.1) of Ehrlich and Kaplan, and none treated initial subgroups: a search of the reports and of the Lean sources for “initial subgroup”, “discrete initial” and “discretely ordered initial” found nothing outside this directory. The paper [1] is cited as background in two reports: the foundations report cites it among the initial-embedding results next to its sign-tree categoricity (`found:sub:signtree`), and the definable-surreals report for the simplicity theory of simplest separators (`dsn:sec:foundations`). The latter’s maximal initial elementary exponential core (`dsn:thm:core`) concerns initial *subfields* of definable surreals; it is related in spirit and neither overlaps nor conflicts with the present groups. The omnific integers used here are those of the Diophantine and set-sized-quotient reports (`odg:def:rings`, `osq:eq:Ozdef`); their discreteness is `odg:prop:units`, and the constant coefficient is the additive map of `osq:lem:ct`. The Diophantine report and the foundations report also cite a different Ehrlich–Kaplan paper, *Surreal ordered exponential fields*, for the initial integer part $\mathbf{Oz} \cap K$ of an initial subfield K (the status note after `odg:thm:floor`, and `found:sub:canonicalexpr`). That statement concerns subrings and integer parts. The groups $H \subseteq \mathbf{Oz}$ here are additive subgroups and $\mathcal{N}_{\alpha,n}$ is not multiplicative, so there is no conflict. No `isg:` statement has a Lean implementation.

Source 02 adds one neighbour. The trigonometry report studies the normalized period groups $\mathcal{I}(\Psi)$ of surcomplex phases, which are $\mathbb{Z}$ -groups for set-sized fields, with the same Jeřábek residue theory: its residue $\text{res}_{\mathcal{I}}$ and maximal divisible subgroup $\mathcal{I}^{\text{div}}$ (`trigonometry:per:sec:defect`) are res_G and G^{div} here, and its profinite defect is that residue modulo $\mathbb{Z}$. By corollary 17.3 a period group is initially realizable exactly when its profinite defect vanishes. There is no conflict: that report’s initiality statements concern the kernel-multiplier ring (`trigonometry:per:thm:robustness`), a subring, and the initial integer part $\mathbf{Oz} \cap F$, not the realizability of period groups. The divisible-kernel lemma is classical in both places. The set-sized quotient report (`osq:eq:Ozdef`) and the Diophantine report use the same Π and ct as proposition 17.1.

20 Further questions and exact boundaries

The construction establishes an additive normal form while deliberately relinquishing several stronger kinds of compatibility. The following questions isolate possible further work without asserting that each is a previously published open problem.

Question 20.1 (Other convex quotients; answered by source 02). Which convex subgroups C of an initial group $A \subseteq \mathbf{No}$ admit an explicit initial realization of A/C by a normal-form transport?

Source 01 handles $C = \mathbb{Z}\varepsilon$. For a larger convex subgroup, one would need to identify the removed exponent region and determine whether its boundary can be relocated without creating unmet dyadic

obligations. A numerical quotient of exponent chains alone would not settle the simplicity condition.

Status: answered, by source 02. All of them. Theorem D and theorem 13.1 realize A/C explicitly: the removed exponent region is the lower segment Γ_C (lemma 11.3), the quotient is represented by the upper projection P_{Γ_C} , a difference of leading truncations (lemma 11.4 and theorem 11.5), and compression of Γ^C creates no unmet dyadic obligation because it reflects right ancestors (27); it is not a numerical quotient of exponent chains (example 12.7). For $C = \mathbb{Z}\varepsilon$ the transport is source 01's (proposition 13.5).

Question 20.2 (Limit iteration; answered by source 02). When an iterated removal of discrete bottom layers continues through infinitely many stages, which hypotheses give a coherent initial realization at a limit stage?

In source 01, finite iteration follows from corollary 8.4; limit iteration requires a separate control of supports and of the changing sign-tree presentations. No commutation with such an unproved transfinite limit is used in source 01.

Status: answered, by source 02 with a corollary of the merge. No hypothesis is needed. The union of the removed layers is a convex subgroup, whose quotient is realized directly from its exponent cut (corollary 13.2), coherently with every earlier stage (corollary 13.3); corollary 13.6 states this for the iteration of corollary 8.4. The required control is supplied by compression, which is defined position by position and never as a limit of successive deletions (remark 12.5).

Question 20.3 (Multiplicative compatibility on restricted classes). For which special initial groups does some normalization to $\mathbf{O}$ also preserve products wherever those products are part of a specified algebraic structure?

The counterexample with $1 \cdot 1$ rules out a general claim for our normalization. Any useful positive result must impose additional conditions on exponent addition and coefficient rescaling, not merely on the order of supports.

Question 20.4 (Formal certification and uniqueness principles). Can the root-relocation proof and the concrete initiality criterion be fully verified in the repository's foundational model, and what natural extra conditions single out this normalization among the possible ordered-group isomorphisms to initial omnific groups?

Our formulas are canonical relative to the given sign model, normal forms, and least positive scale. They are not asserted to be invariant under arbitrary abstract group isomorphisms. The uniqueness in Theorem B concerns the inverse under the fixed map, not uniqueness of all possible initial embeddings.

The next four questions are source 02's. They mark extensions of the proved statements rather than prerequisites left unproved in their proofs, and they are not asserted to be previously published open problems (the second is related to the published Question 1 / 8.1 of Ehrlich and Kaplan, section 18.2).

Question 20.5 (Intrinsic characterization). Which initially realizable ordered groups are characterized by intrinsic conditions on all convex factors and the splitting of convex extensions, without first choosing a Hahn presentation?

Question 20.6 (Intermediate theories). Can the nonsplit convex-extension obstruction be arranged inside a set-sized model of every proper intermediate theory appearing in the Ehrlich–Kaplan optimality question?

Question 20.7 (Arithmetic fragments). What is the weakest arithmetic fragment whose nonstandard models have ordered additive group completions excluded by the factorial residue type? Which analogous computable residue types yield sharper bounds?

Question 20.8 (Compression and multiplication). Under what additional compatibility conditions does exponent-tree compression preserve a specified multiplication or a specified action of a coefficient domain? The unrestricted additive result provides no such compatibility automatically.

This is the counterpart for compression of [question 20.3](#), which asks the same for the normalization $N_{\alpha,n}$.

20.1 Non-claims, collected

The limitations stated at their places in the text, in the shipped source audit and in the shipped provenance record of source 02 are collected here, so that none is lost. Each stays in force where it was stated. Items (N1)–(N15) are source 01’s; in them “the manuscript” means source 01. Items (N16)–(N29) are source 02’s.

- (N1) *Status*. The manuscript is unrefereed. It is a proposed solution with full proposed proofs, not an independently certified resolution; neither correctness nor novelty is certified, and no priority is claimed. The placement review ([section 19.5](#)) is not a refereeing.
- (N2) *Literature*. The literature and repository searches were targeted, not exhaustive. A negative search result is limited evidence and does not exclude an unindexed paper, an unpublished proof or a reformulation known to specialists. That a question was posed in 2015 and 2018 does not establish that it was still unresolved on the review date.
- (N3) *Imported criterion*. [Imported fact 2.2](#) is imported, not re-proved, by the manuscript or at placement. No proof here replaces it by the weaker assertion that initiality of the exponent class alone suffices.
- (N4) *Not claimed as new*. The two-level example and its map, the initial-Hahn-subgroup characterization, the classification of initial real groups ([imported fact 2.3](#)) and Conway normal forms are imported and cited. The extremal, quotient and surcomplex statements are derived refinements of the construction.
- (N5) *Not a simplicity isomorphism*. $N_{\alpha,n}$ is an ordered-group isomorphism onto an initial image. It need not preserve the simplicity relation ([remark 6.3](#)); it is not multiplicative, does not preserve norms, and does not fix ordinary constants: $N_{\alpha,n}(1) \neq 1$ in general ([section 10](#)).
- (N6) *No classification*. No classification of all ordered abelian groups admitting initial embeddings is claimed, and the initial-monoid problem ([Question 3](#) of [1], [Question 9.2](#) of the journal version) is not addressed.
- (N7) *Exponent maps*. Θ_α is not a homomorphism of exponent groups, and [\(15\)](#) is not preservation of a multiplicative valuation ([remark 5.4](#)). [Proposition 4.8](#) bounds exponent lengths, not the birthday of a transformed normal form.
- (N8) *No internal summation*. The strong-sum statements concern the ambient Hahn spaces; they give no internal summation on a subgroup A ([remark 5.3](#)), and [lemma 2.4](#) does not license arbitrary infinite subseries.
- (N9) *Invariance and uniqueness*. The ordinal α is not an invariant of the abstract ordered group. The uniqueness in [Theorem B](#) is relative to the fixed map, not uniqueness of all initial embeddings, and the formulas are not claimed invariant under abstract isomorphisms.
- (N10) *Quotients*. It is not claimed that every convex quotient of an initial group has an initial realization. Iteration is covered only at finite stages ([corollary 8.4](#)), with no claim at a limit stage, and [corollary 8.6](#) is a relative criterion that does not decide realizability of every quotient. (This is a statement about source 01. Source 02 proves both missing statements: [Theorem D](#) and [corollaries 13.2](#) and [13.6](#). The relative criterion remains relative; [Theorem E](#) decides the case of $\mathbb{Z}$ -groups.)
- (N11) *Surcomplex scope*. [Theorem 9.2](#) is Gaussian-linear and coordinatewise initial. It gives no field isomorphism, no canonical sign-tree or simplicity relation on $\mathbf{No}[i]$, no preservation of the

modulus, and no compatibility with multiplication by general surcomplex scalars. Corollary 9.3 transports the right-hand side as well and does not apply to equations with ordinary constants held fixed, to nonlinear polynomials, to arbitrary surreal scalar matrices, or to norms and inner products.

- (N12) *Finite checks.* The finite checks do not verify transfinite concatenation or limit cases, arbitrary set-supported Hahn sums, the imported criterion, or the absence of a transfinite proof gap; the written proofs carry the general claims. Compiling and rendering the document establish its integrity, not its correctness.
- (N13) *No Lean.* No new Lean formalization and no Lean build are claimed. The proposed formalization units of section 19 are not existing declarations, and a checked conditional theorem using the imported criterion would not by itself be a checked theorem about every initial subgroup of **No**.
- (N14) *Repository review.* The manuscript’s repository review was targeted: it did not review every report or archive, it treated truncated metadata as neither a complete inventory nor a proof of absence, it took no repository description as a certificate, and it promoted no repository result from pending to proved. The argument relies on no repository manuscript, and no repository Lean build was run for it.
- (N15) *Sources used only for context.* Bagayoko and van der Hoeven’s sign-sequence conventions are context, not a source of the normalization or of its range theorem. The questions of this section are not asserted to be previously published open problems, and $d(H) = \mathbf{On}$ is a formal marker, not a surreal number or a set ordinal.
- (N16) *Status of source 02.* The convex-factor theorem and its applications are proposed contributions with complete proofs. Neither priority nor independent referee certification is claimed, no Lean formalization or Lean build is claimed, and the placement review (section 19.5) is not a refereeing.
- (N17) *Credited, not new.* The Ehrlich–Kaplan criterion, Ehrlich’s divisible embedding (imported fact 11.1), Jeřábek’s residue theory and rigidity mechanism, and $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ are classical and credited; no new computation of Ext is claimed, and the existence of rigid Presburger models is not claimed as a new phenomenon. The results of source 01 that source 02 uses (the suspension, the relative splitting criterion) are not claimed again as new.
- (N18) *No classification.* There is no classification of initially realizable groups. Convex splitting is necessary, and its sufficiency is not proved (section 18).
- (N19) *Optimality question.* Source 02 does not resolve the Ehrlich–Kaplan set-model optimality question (Question 1 of [1], Question 8.1 of the journal version) and gives no construction uniform over the intermediate theories (section 18.2 and remark 18.1). It answers none of Ehrlich and Kaplan’s questions.
- (N20) *Realization-dependent splitting.* The splitting of theorem 11.5 depends on the chosen realization and is not invariant under abstract automorphisms; only the Presburger split case has the uniqueness of proposition 14.3.
- (N21) *Realizability, not literal initiality.* The results concern initial *realizability*, not literal initiality of C or of a complement in its original position.
- (N22) *No Hahn product, no completion.* An initial group is not identified with the full Hahn product of its factors, and no closure under ambient strong sums is asserted (remarks 11.2 and 13.4).
- (N23) *Additive only.* The results are additive and module-theoretic. There is no multiplicative, field, ring, differential or exponential equivalence; the compression need not respect addition of exponents.

- (N24) *Surcomplex scope of source 02.* The rectangular notion is coordinate-dependent; no canonical sign tree on $\mathbf{No}[i]$ is postulated.
- (N25) *Finite checks of source 02.* The finite regression does not verify the Ehrlich–Kaplan theorem, arbitrary ordinals, class comprehension, completeness of Presburger arithmetic, recursive saturation or the Peano-model argument; the successful finite retractions are not a global splitting (remark 15.4). The delivered build and rendering checks concern presentation, not correctness.
- (N26) *Arithmetic fragment.* The fragment of arithmetic used for theorem 16.6 is not optimized.
- (N27) *Repository review of source 02.* Its review, at pin 9693b28, was targeted: it was not a claim-by-claim audit of the repository, and no local Lean build was run.
- (N28) *Predecessor citation.* Source 02 cited source 01 without inventing a repository path for it; the current location (section 19) was supplied at this write.
- (N29) *Finite solvability.* Solvability of every finite retraction system is not a global splitting (remark 15.4).

The statements marked [merge] add no claim of priority; they are proved from the two sources and, for corollary 17.3, from statements of the trigonometry report cited by label, whose own non-claims remain in force there.

21 Conclusion

The obstruction in the Ehrlich–Kaplan question is not the numerical size of the least positive element. Scalar multiplication removes that size immediately. The obstruction is the interaction of the exponent tree with coefficient groups forced by right ancestors.

The explicit map Θ_α moves the minimum exponent to zero and preserves all right-ancestor relations away from it. The coefficient map then normalizes the discrete bottom group without disturbing the higher groups. This gives the proposed affirmative proof. The relations lost at the minimum are exactly those recorded by the dyadic spine, which yields a necessary-and-sufficient image criterion rather than only a sufficient construction.

The resulting theory is additive but strong: it transports arbitrary set-supported normal forms without changing their support order types, commutes with strong sums, identifies extremal initial groups at each least positive scale, and produces initial models of the bottom cyclic quotients and rectangular surcomplex modules. The remaining obligations are independent mathematical review, a more exhaustive priority check, and a complete formal verification of the identified dependency chain.

Source 02’s compression construction makes convex-factor closure a concrete normal-form theorem: retain an exponent interval, delete precisely the ancestor positions outside it, and invoke the initial Hahn criterion. It is coherent for nested intervals and works at arbitrary transfinite cuts; for the bottom cyclic layer it is source 01’s $\rho \circ \Theta_\alpha$. At the same time, normal-form projection forces all convex extensions inside an initial realization to split. In additive arithmetic this yields a sharp boundary: a Presburger group has an initial surreal realization exactly when each element’s coherent standard residues are ordinary integers. The explicit factorial family places continuum many countable rigid models on the other side of that boundary. Dense examples and omitted-type arguments show that the distinction survives beyond discreteness and cannot be captured by first-order theory alone. None of this is a certified resolution of the full classification problem or of Ehrlich and Kaplan’s optimality question.

A A compact proof for independent checking

This appendix compresses the main existence argument without changing its hypotheses. The full case analysis and support arguments are in the body of the paper.

Let A be a discrete initial subgroup. Its least positive member has a single normal-form term: otherwise a nonzero tail in A would be a smaller positive element after taking absolute value. Write that term as $c\omega^p$. The exponent class has minimum p , which, being the minimum of an initial tree, is all-minus, say $p = -\alpha$. The bottom real coefficient group is discrete and initial, hence $c\mathbb{Z}$ with $c = 2^{-n}$.

On $[p, \infty)$ define

$$p \mapsto 0, \quad p \frown \langle + \rangle \frown u \mapsto \langle + \rangle \frown p \frown u, \quad x \mapsto \langle + \rangle \frown x \quad (p \not\leq x).$$

The inverse removes the initial plus and reinserts a plus immediately after the block p whenever the remaining word starts with p . The two maps are increasing inverse bijections. For $x > p$,

$$\text{Pred}(\Theta_\alpha x) = \Theta_\alpha[\text{Pred}(x) \cup \{p\}], \quad \text{R}_s(\Theta_\alpha x) = \Theta_\alpha[\text{R}_s(x)].$$

Therefore an initial exponent class with minimum p maps to an initial one with minimum zero, and every new right-ancestor obligation comes from an old one above p .

Transport each nonbottom term $r\omega^\gamma$ to $r\omega^{\Theta_\alpha \gamma}$, and transport the bottom term $r\omega^p$ to r/c . This is a real-linear order-isomorphism on the ambient Hahn spaces. Its image of A is truncation closed and cross-sectional, has coefficient group $\mathbb{Z}$ at zero, and has unchanged initial real coefficient groups at positive exponents. The right-ancestor identity preserves all required dyadic containments. By the concrete Ehrlich–Kaplan criterion, the image is initial. It lies in **Oz** because all exponents are nonnegative and the constant coefficient is integral.

For the inverse classification, only the ancestors of the old minimum were discarded. They are $-\beta$ for $\beta < \alpha$, and their images are $q_\beta = \langle + \rangle \frown \langle - \rangle^\beta$. Initial inversion requires those exponents, while the right-ancestor condition at p requires all their dyadic coefficients. Together these requirements are exactly $K_\alpha \subseteq H$.

B Proof-review checklist

An independent review can focus on the following specific potential failure points rather than checking experimental output as though it were a proof.

Issue	Location and required check
Concrete versus existential initiality	Imported fact 2.2: the imported sufficiency statement must apply to the specified canonical image, not only to an unspecified isomorphic copy.
Minimum exponent	Proposition 3.2: truncation and subtraction isolate a nonzero tail; its absolute value is smaller than the positive leading monomial.
Limit sign lengths	Lemma 4.2 and theorem 4.4: concatenated tails use ordinal interval order types. No subtraction of ordinal lengths or removal of a last limit sign is permitted.
Exceptional old minimum	Theorem 4.4 and remark 4.5: the right-ancestor equality excludes $x = p$. Its failure is used, not ignored, in the inverse classification.
Coefficient group scaling	Lemma 6.1: only the bottom coefficient is rescaled; all right-ancestor obligations in the target are at unchanged groups.
Infinite subgroup membership	Section 5: transport the actual subgroup. Never replace it with all normal forms allowed by its coefficient groups.
Sharp inverse criterion	Theorem B: inverse exponent initiality is not enough; the missing minimum's ancestors must carry all dyadic coefficients.
Quotient order	Lemma 8.1 and proposition 8.3: higher nonzero terms dominate the cyclic bottom group, and positive-root deletion supplies an initial image.
Class-sized scope	All supports and summation indices are sets; correspondences of class-sized groups are expressed with class parameters.
Surcomplex scope	Theorem 9.2: the conclusion is Gaussian-linear and coordinatewise initial, not a field isomorphism or a new canonical complex simplicity relation.
<i>Source 02 (convex factors).</i>	
Actual normal-form image	Imported fact 2.2: its sufficiency part must apply to the actual transported image $\mathcal{F}_E(G)$, not just to an abstract copy.
Restriction stays in G	Lemma 11.4: P_E is the difference of two leading truncations, so no infinite subseries is used.
Meet retention	Theorem 12.4: compression must retain the sign at the meet of an incomparable pair; convexity puts the meet in E (lemma 12.2 and example 12.8).
Every target right ancestor	Theorems 12.4 and 13.1: every target right ancestor must be accounted for, not merely every source ancestor; this is the equality (27).
Residue sign	Proposition 15.1: the residue of x_1 is $-\zeta$, not ζ ; factorial moduli are cofinal for divisibility.
Divisible complement	Proposition 14.3: a complement to $\mathbb{Z}e$ is divisible because it is isomorphic to the quotient, not because every torsion-free subgroup is divisible.
Noninteger parameter	Lemma 15.3: ζ_* is shown to be neither a nonnegative nor a negative ordinary integer.
Internal sums	Theorem 16.6: the sum in a nonstandard Peano model is an internally coded finite sum, not an external infinite sum in G .

Issue	Location and required check
Merge statements	Proposition 13.5 and corollaries 13.6 and 17.3: the identification with $\rho \circ \Theta_\alpha$, the one-point layers of the iteration, and the order of $I/\mathbb{Z} \cong \Pi_F$.

C The core convex-factor proof in one place

This appendix is source 02's. For independent checking, the logical chain can be compressed without changing its hypotheses. Let $G \subseteq \mathbf{No}$ be initial and let $C \subseteq C'$ be convex. The active exponents form an initial class Γ . Convexity says that membership of a nonzero element in C or C' is determined exactly by whether its leading monomial belongs to that subgroup. Thus each has a lower exponent segment, Γ_C and $\Gamma_{C'}$.

The interval $E = \Gamma_{C'} \setminus \Gamma_C$ represents C'/C : coefficient restriction to E stays in G , because it is a difference of two leading truncations. The kernel on C' is C , and the leading sign gives the quotient order.

Any two incomparable elements of E have their common prefix between them numerically. It is in Γ by initiality and in E by convexity. Thus E is meet closed. In each $\gamma \in E$, retain the signs at precisely those positions β for which $\gamma \upharpoonright \beta \in E$. Index them by their ordinal order type. A source prefix in E becomes a target prefix; incomparable points keep opposite signs at their retained meet. Every target prefix comes from one of the retained source prefixes. Hence the map is an order-isomorphism onto an initial exponent class and preserves and reflects right ancestry.

Reindex the normal forms supported in E by this map. The coefficient groups do not change, truncation closure and cross-sectionality persist, and every target dyadic right-ancestor requirement is one of the original requirements. The Ehrlich–Kaplan criterion makes this actual image an initial subgroup of $\mathbf{No}$. This proves the convex-factor assertion. Upper-exponent restriction simultaneously supplies the split section for each convex subgroup sequence.

References

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- [2] P. Ehrlich and E. Kaplan, *Number systems with simplicity hierarchies: a generalization of Conway's theory of surreal numbers II*, The Journal of Symbolic Logic **83** (2018), no. 2, 617–633. <https://doi.org/10.1017/jsl.2017.9>. Publisher metadata checked; the arXiv version above supplies the cited internal numbering. The journal numbering given in section 1 (Lemma 5.1, Theorem 5.1, Proposition 5.2, Theorem 8.1, Questions 8.1, 9.1 and 9.2) was read, at placement, in the authors' text of this version hosted on the second author's web page; the publisher's typeset pages were not compared.
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- [11] N. Strickland, *Algebraic theory of abelian groups*, <https://arxiv.org/abs/2001.10469> (2020), especially Section 8 and Example 8.27. Cited by source 02 for $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$.
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